tropical: ThreeSumDecision

2026-10-26T03:54:11

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `096095567af4`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the rank-based path: more than 20% of values within 0.01 of a bound (0, 1). Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.01455	[0.0107, 0.0181]	0.00479	5	0
adaptive	pointer	0.01	0.01514	[0.0119, 0.0185]	0.0042	5	0
adaptive	pointer	0.1	0.01222	[0.0108, 0.0136]	0.00182	5	0
adaptive	pointer	1	0.01614	[0.0122, 0.0206]	0.00537	5	0
adaptive	pointer	10	0.01459	[0.0111, 0.018]	0.00447	5	0
adaptive	pointer	100	0.0133	[0.0113, 0.0163]	0.00332	5	0
adaptive	pointer	1000	0.01488	[0.0112, 0.0185]	0.00468	5	0
adaptive	pointer	1e+04	0.01512	[0.0124, 0.0178]	0.00357	5	0
adaptive	seq2seq	0	0	[0, 0]	0	5	0
adaptive	seq2seq	0.01	0	[0, 0]	0	5	0
adaptive	seq2seq	0.1	0	[0, 0]	0	5	0
adaptive	seq2seq	1	0	[0, 0]	0	5	0
adaptive	seq2seq	10	0	[0, 0]	0	5	0
adaptive	seq2seq	100	0	[0, 0]	0	5	0
adaptive	seq2seq	1000	0	[0, 0]	0	5	0
adaptive	seq2seq	1e+04	3.743×10^{-5}	[0, 1.12×10^{-4}]	8.37×10^{-5}	5	0
tropical	pointer	0	0.03371	[0.0163, 0.0545]	0.0256	5	0
tropical	pointer	0.01	0.03993	[0.0202, 0.0627]	0.0266	5	0
tropical	pointer	0.1	0.04958	[0.0379, 0.0621]	0.0162	5	0
tropical	pointer	1	0.03209	[0.0109, 0.0556]	0.0287	5	0
tropical	pointer	10	0.04421	[0.0221, 0.0717]	0.0328	5	0
tropical	pointer	100	0.03821	[0.0157, 0.0607]	0.0303	5	0
tropical	pointer	1000	0.04064	[0.0214, 0.0604]	0.026	5	0
tropical	pointer	1e+04	0.03573	[0.0167, 0.0539]	0.0248	5	0
tropical	seq2seq	0	0	[0, 0]	0	5	0
tropical	seq2seq	0.01	0.003202	[0, 0.00953]	0.00706	5	0
tropical	seq2seq	0.1	3.299×10^{-4}	[0, 9.90×10^{-4}]	7.38×10^{-4}	5	0
tropical	seq2seq	1	0.001692	[0, 0.005]	0.00368	5	0
tropical	seq2seq	10	0.002665	[0, 0.008]	0.00596	5	0
tropical	seq2seq	100	0	[0, 0]	0	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0	[0, 0]	0	5	0
tropical	seq2seq	1e+04	0	[0, 0]	0	5	0
vanilla	pointer	0	0.0348	[0.0212, 0.0484]	0.0173	5	0
vanilla	pointer	0.01	0.03937	[0.0299, 0.0504]	0.0131	5	0
vanilla	pointer	0.1	0.04106	[0.0308, 0.0515]	0.0133	5	0
vanilla	pointer	1	0.04055	[0.0305, 0.05]	0.0131	5	0
vanilla	pointer	10	0.04065	[0.0305, 0.051]	0.0127	5	0
vanilla	pointer	100	0.03928	[0.0294, 0.0505]	0.0132	5	0
vanilla	pointer	1000	0.03868	[0.0286, 0.0491]	0.0134	5	0
vanilla	pointer	1e+04	0.03897	[0.0291, 0.0508]	0.0137	5	0
vanilla	seq2seq	0	0	[0, 0]	0	5	0
vanilla	seq2seq	0.01	1.100×10^{-4}	$[0, 3.30 \times 10^{-4}]$	2.46×10^{-4}	5	0
vanilla	seq2seq	0.1	2.967×10^{-4}	$[0, 7.80 \times 10^{-4}]$	5.67×10^{-4}	5	0
vanilla	seq2seq	1	3.714×10^{-5}	$[0, 1.11 \times 10^{-4}]$	8.31×10^{-5}	5	0
vanilla	seq2seq	10	0	[0, 0]	0	5	0
vanilla	seq2seq	100	0	[0, 0]	0	5	0
vanilla	seq2seq	1000	2.222×10^{-4}	$[0, 5.93 \times 10^{-4}]$	4.02×10^{-4}	5	0
vanilla	seq2seq	1e+04	0	[0, 0]	0	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-5.151×10^{-4}	[-0.00126, 0.00307]	1.0000
adaptive	pointer	0.1	-0.002643	[-0.00622, 0.00144]	1.0000
adaptive	pointer	1	0.001255	$[-6.43 \times 10^{-4}, 0.004]$	1.0000
adaptive	pointer	10	-2.778×10^{-5}	[-0.00187, 0.0024]	1.0000
adaptive	pointer	100	-1.832×10^{-4}	[-0.00439, 0.00133]	1.0000
adaptive	pointer	1000	5.554×10^{-4}	[-0.00131, 0.00219]	1.0000
adaptive	pointer	1e+04	1.844×10^{-4}	$[-9.32 \times 10^{-4}, 0.00256]$	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	$[0, 1.12 \times 10^{-4}]$	1.0000
tropical	pointer	0.01	0.006365	[-0.0034, 0.0168]	1.0000
tropical	pointer	0.1	0.01179	[0.00507, 0.0277]	0.8750
tropical	pointer	1	-8.919×10^{-5}	[-0.0084, 0.00485]	1.0000
tropical	pointer	10	0.009476	[0.00413, 0.0161]	0.8750
tropical	pointer	100	0.006045	[-0.00193, 0.0117]	1.0000
tropical	pointer	1000	0.01222	[-0.00282, 0.0157]	0.9375
tropical	pointer	1e+04	-0.001735	[-0.00838, 0.0149]	1.0000
tropical	seq2seq	0.01	0	[0, 0.00953]	1.0000
tropical	seq2seq	0.1	0	$[0, 9.90 \times 10^{-4}]$	1.0000
tropical	seq2seq	1	0	[0, 0.005]	1.0000
tropical	seq2seq	10	0	[0, 0.008]	1.0000
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0.002964	[-0.00163, 0.0114]	1.0000
vanilla	pointer	0.1	0.004343	[-0.00125, 0.015]	1.0000

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	1	0.0033	$[-2.38 \times 10^{-4}, 0.0121]$	1.0000
vanilla	pointer	10	0.003081	$[-3.83 \times 10^{-4}, 0.0124]$	1.0000
vanilla	pointer	100	0.003157	[-0.00167, 0.0108]	1.0000
vanilla	pointer	1000	0.002459	[-0.00209, 0.0108]	1.0000
vanilla	pointer	1e+04	0.004002	[-0.00172, 0.0103]	1.0000
vanilla	seq2seq	0.01	0	$[0, 3.30 \times 10^{-4}]$	1.0000
vanilla	seq2seq	0.1	0	$[0, 7.80 \times 10^{-4}]$	1.0000
vanilla	seq2seq	1	0	$[0, 1.11 \times 10^{-4}]$	1.0000
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	$[0, 5.57 \times 10^{-4}]$	1.0000
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 3: `ood_len_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.01273	[-0.163, 0.0949]	0.9421	0.00571	0.9804
adaptive	seq2seq	0.2572	[0.257, 0.459]	0.1357	0.0571	0.2815
tropical	pointer	-0.05234	[-0.281, 0.316]	0.7653	-0.0324	0.8144
tropical	seq2seq	-0.3769	[-0.622, -0.25]	0.0256	-0.189	0.0204
vanilla	pointer	-0.1245	[-0.403, -0.0709]	0.4762	-0.0971	0.4695
vanilla	seq2seq	-0.1529	[-0.307, -0.0539]	0.3805	-0.0781	0.3813

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.001095	[-0.0341, 0.0319]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	5.569×10^{-4}	[-0.0312, 0.0323]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.008519	[-0.0221, 0.0391]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.008456	[-0.0397, 0.0228]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.003557	[-0.0365, 0.0436]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.001069	[-0.0354, 0.0332]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.001951	[-0.0256, 0.0295]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.003236	[-0.0304, 0.024]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.003092	[-0.00536, 0.0115]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	3.321×10^{-5}	[-0.00118, 0.00125]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.001655	[-0.00294, 0.00625]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.002665	[-0.00473, 0.0101]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-2.222×10^{-4}	$[-7.22 \times 10^{-4}, 2.77 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.02026	[-0.0444, 0.00385]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.02423	[-0.0388, -0.00966]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.02884	[-0.0452, -0.0125]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.02441	[-0.0404, -0.00846]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.02606	[-0.0414, -0.0107]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.02597	[-0.0406, -0.0113]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.0238	[-0.0391, -0.00851]	0.8125
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.02386	[-0.0395, -0.00821]	0.8125
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-1.100×10^{-4}	$[-4.15 \times 10^{-4}, 1.95 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-2.967×10^{-4}	$[-0.001, 4.07 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-3.714×10^{-5}	$[-1.40 \times 10^{-4}, 6.60 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-2.222×10^{-4}	$[-7.22 \times 10^{-4}, 2.77 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	3.743×10^{-5}	$[-6.65 \times 10^{-5}, 1.41 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01455	[0.0086, 0.0205]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01514	[0.00993, 0.0204]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01222	[0.00996, 0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01614	[0.00947, 0.0228]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01459	[0.00904, 0.0201]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0133	[0.00918, 0.0174]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01488	[0.00906, 0.0207]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01508	[0.0107, 0.0194]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.03371	[0.00191, 0.0655]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.03673	[0.00122, 0.0722]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.04925	[0.0284, 0.07]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.0304	[-0.0073, 0.0681]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.04154	[-0.00428, 0.0874]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.03821	$[5.78 \times 10^{-4}, 0.0758]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.04064	[0.0084, 0.0729]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.03573	[0.005, 0.0665]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0348	[0.0134, 0.0562]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.03926	[0.0229, 0.0556]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.04076	[0.0242, 0.0573]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.04051	[0.0243, 0.0568]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.04065	[0.0248, 0.0565]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.03928	[0.0229, 0.0557]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.03846	[0.0217, 0.0552]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03897	[0.0219, 0.056]	1.0000

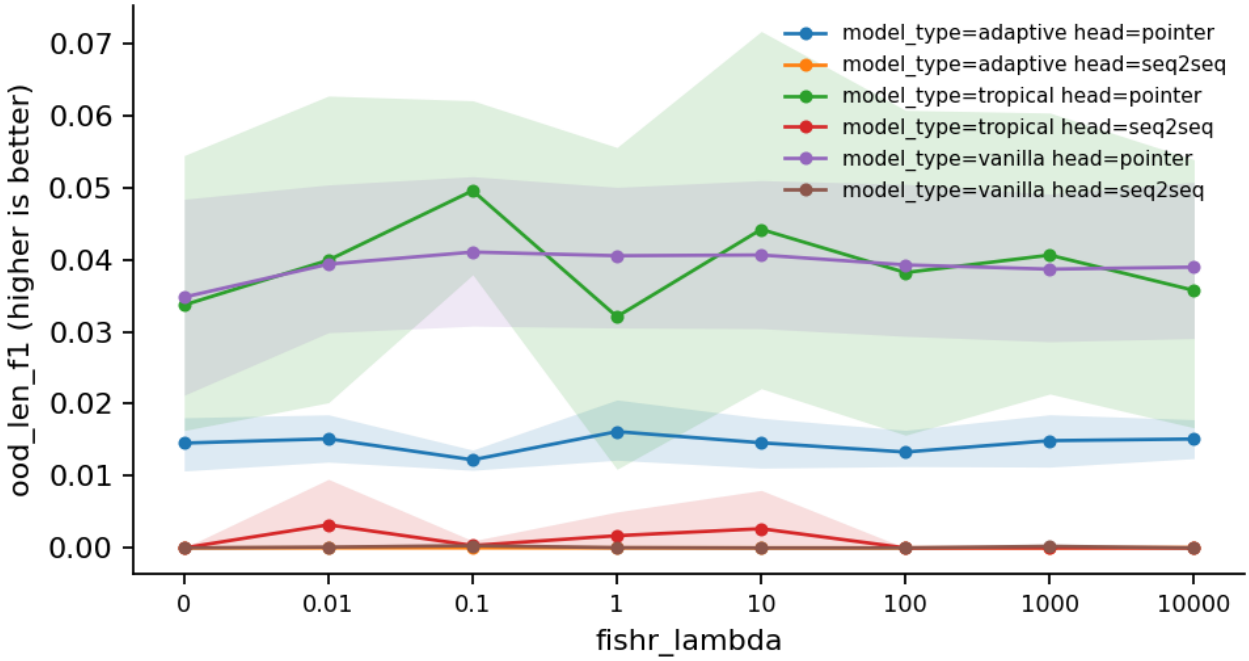


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

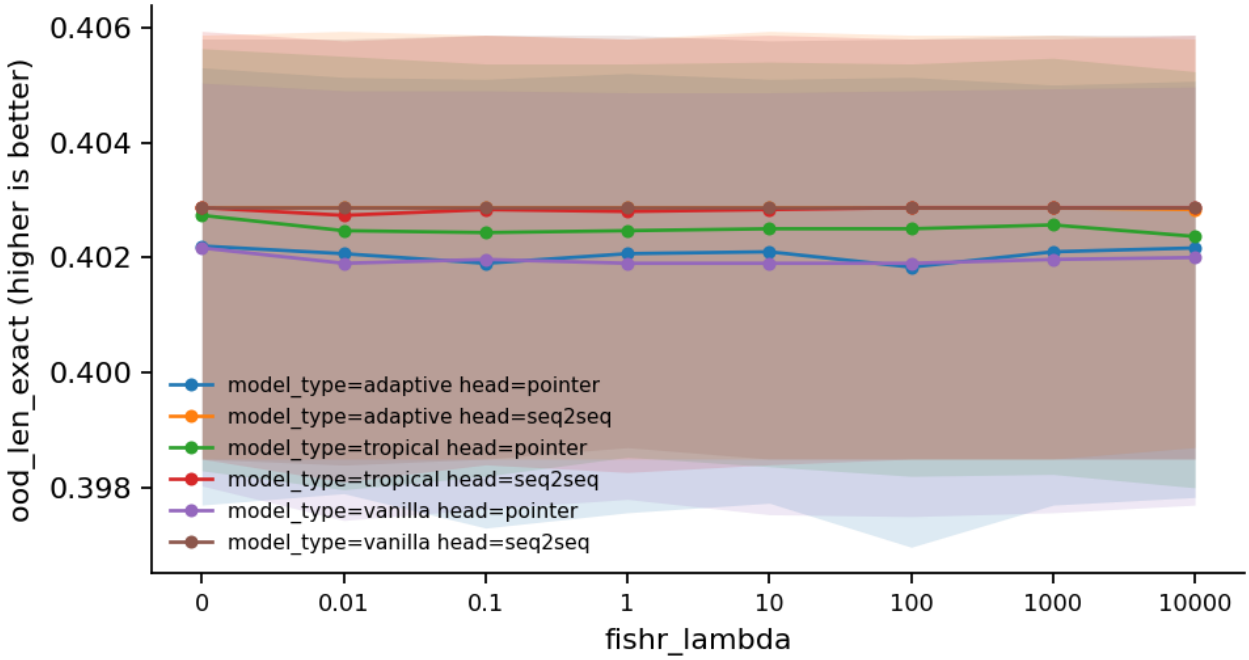


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4022	[0.398, 0.405]	0.00506	5	0
adaptive	pointer	0.01	0.4021	[0.398, 0.405]	0.00499	5	0
adaptive	pointer	0.1	0.4019	[0.397, 0.405]	0.00516	5	0
adaptive	pointer	1	0.4021	[0.398, 0.405]	0.00509	5	0
adaptive	pointer	10	0.4021	[0.398, 0.405]	0.00491	5	0
adaptive	pointer	100	0.4018	[0.397, 0.405]	0.00542	5	0
adaptive	pointer	1000	0.4021	[0.398, 0.405]	0.00498	5	0
adaptive	pointer	1e+04	0.4022	[0.398, 0.405]	0.0049	5	0
adaptive	seq2seq	0	0.4029	[0.398, 0.406]	0.00488	5	0
adaptive	seq2seq	0.01	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	0.1	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	1	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	10	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	100	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	1000	0.4029	[0.399, 0.406]	0.00488	5	0
adaptive	seq2seq	1e+04	0.4028	[0.399, 0.406]	0.00486	5	0
tropical	pointer	0	0.4027	[0.398, 0.406]	0.00483	5	0
tropical	pointer	0.01	0.4025	[0.398, 0.405]	0.00502	5	0
tropical	pointer	0.1	0.4024	[0.398, 0.405]	0.00478	5	0
tropical	pointer	1	0.4025	[0.399, 0.405]	0.00485	5	0
tropical	pointer	10	0.4025	[0.398, 0.405]	0.00468	5	0
tropical	pointer	100	0.4025	[0.398, 0.405]	0.00475	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.4026	[0.398, 0.405]	0.00483	5	0
tropical	pointer	1e+04	0.4024	[0.398, 0.405]	0.00485	5	0
tropical	seq2seq	0	0.4029	[0.399, 0.406]	0.00488	5	0
tropical	seq2seq	0.01	0.4027	[0.398, 0.406]	0.00517	5	0
tropical	seq2seq	0.1	0.4028	[0.398, 0.406]	0.00496	5	0
tropical	seq2seq	1	0.4028	[0.398, 0.406]	0.00493	5	0
tropical	seq2seq	10	0.4028	[0.398, 0.406]	0.00496	5	0
tropical	seq2seq	100	0.4029	[0.399, 0.406]	0.00488	5	0
tropical	seq2seq	1000	0.4029	[0.399, 0.406]	0.00488	5	0
tropical	seq2seq	1e+04	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	pointer	0	0.4022	[0.398, 0.405]	0.00466	5	0
vanilla	pointer	0.01	0.4019	[0.397, 0.405]	0.00504	5	0
vanilla	pointer	0.1	0.402	[0.398, 0.405]	0.00489	5	0
vanilla	pointer	1	0.4019	[0.398, 0.405]	0.00488	5	0
vanilla	pointer	10	0.4019	[0.398, 0.405]	0.00489	5	0
vanilla	pointer	100	0.4019	[0.397, 0.405]	0.00496	5	0
vanilla	pointer	1000	0.402	[0.398, 0.405]	0.0049	5	0
vanilla	pointer	1e+04	0.402	[0.398, 0.405]	0.00485	5	0
vanilla	seq2seq	0	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	0.01	0.4029	[0.398, 0.406]	0.00488	5	0
vanilla	seq2seq	0.1	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	1	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	10	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	100	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	1000	0.4029	[0.399, 0.406]	0.00488	5	0
vanilla	seq2seq	1e+04	0.4029	[0.399, 0.406]	0.00488	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-1.333×10^{-4}	[-0.00898, 0.00872]	1.0000
adaptive	pointer	0.1	-3.000×10^{-4}	[-0.00915, 0.00855]	1.0000
adaptive	pointer	1	-1.333×10^{-4}	[-0.00898, 0.00872]	1.0000
adaptive	pointer	10	-1.000×10^{-4}	[-0.00895, 0.00875]	1.0000
adaptive	pointer	100	-3.667×10^{-4}	[-0.00922, 0.00848]	1.0000
adaptive	pointer	1000	-1.000×10^{-4}	[-0.00895, 0.00875]	1.0000
adaptive	pointer	1e+04	-3.333×10^{-5}	[-0.00888, 0.00882]	1.0000
adaptive	seq2seq	0.01	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	0.1	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	1	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	10	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	100	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	1000	0	[-0.00853, 0.00853]	1.0000
adaptive	seq2seq	1e+04	-3.333×10^{-5}	[-0.00856, 0.00849]	1.0000
tropical	pointer	0.01	-2.667×10^{-4}	[-0.00869, 0.00816]	1.0000
tropical	pointer	0.1	-3.000×10^{-4}	[-0.00873, 0.00813]	1.0000
tropical	pointer	1	-2.667×10^{-4}	[-0.00869, 0.00816]	1.0000
tropical	pointer	10	-2.333×10^{-4}	[-0.00866, 0.00819]	1.0000
tropical	pointer	100	-2.333×10^{-4}	[-0.00866, 0.00819]	1.0000
tropical	pointer	1000	-1.667×10^{-4}	[-0.00859, 0.00826]	1.0000
tropical	pointer	1e+04	-3.667×10^{-4}	[-0.00879, 0.00806]	1.0000
tropical	seq2seq	0.01	-1.333×10^{-4}	[-0.00877, 0.0085]	1.0000
tropical	seq2seq	0.1	-3.333×10^{-5}	[-0.00867, 0.0086]	1.0000

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1	-6.667×10^{-5}	[-0.0087, 0.00857]	1.0000
tropical	seq2seq	10	-3.333×10^{-5}	[-0.00867, 0.0086]	1.0000
tropical	seq2seq	100	0	[-0.00864, 0.00864]	1.0000
tropical	seq2seq	1000	0	[-0.00864, 0.00864]	1.0000
tropical	seq2seq	1e+04	0	[-0.00864, 0.00864]	1.0000
vanilla	pointer	0.01	-2.667×10^{-4}	[-0.0088, 0.00827]	1.0000
vanilla	pointer	0.1	-2.000×10^{-4}	[-0.00873, 0.00833]	1.0000
vanilla	pointer	1	-2.667×10^{-4}	[-0.0088, 0.00827]	1.0000
vanilla	pointer	10	-2.667×10^{-4}	[-0.0088, 0.00827]	1.0000
vanilla	pointer	100	-2.667×10^{-4}	[-0.0088, 0.00827]	1.0000
vanilla	pointer	1000	-2.000×10^{-4}	[-0.00873, 0.00833]	1.0000
vanilla	pointer	1e+04	-1.667×10^{-4}	[-0.0087, 0.00837]	1.0000
vanilla	seq2seq	0.01	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	0.1	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	1	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	10	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	100	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	1000	0	[-0.00853, 0.00853]	1.0000
vanilla	seq2seq	1e+04	0	[-0.00853, 0.00853]	1.0000

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.667×10^{-5}	$[-8.33 \times 10^{-6}, 4.88 \times 10^{-5}]$	0.9666	0.0171	0.8996
adaptive	seq2seq	-3.571×10^{-6}	$[-1.07 \times 10^{-5}, 0]$	0.9926	-0.0114	0.9300
tropical	pointer	3.965×10^{-19}	$[-2.98 \times 10^{-5}, 2.98 \times 10^{-5}]$	1.0000	-0.0133	0.9232
tropical	seq2seq	1.905×10^{-5}	$[0, 5.48 \times 10^{-5}]$	0.9609	0.0324	0.8164
vanilla	pointer	1.071×10^{-5}	$[-1.07 \times 10^{-5}, 3.21 \times 10^{-5}]$	0.9778	0.0343	0.7954
vanilla	seq2seq	0	[0, 0]	1.0000	0	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	5.667×10^{-4}	$[9.02 \times 10^{-5}, 0.00104]$	0.3285
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	5.667×10^{-4}	$[-8.12 \times 10^{-5}, 0.00121]$	0.6618
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	4.667×10^{-4}	$[-1.44 \times 10^{-4}, 0.00108]$	0.7073
model_type: tropical - vanilla head=pointer fishr_lambda=1	5.667×10^{-4}	$[-2.82 \times 10^{-4}, 0.00141]$	0.8233
model_type: tropical - vanilla head=pointer fishr_lambda=10	6.000×10^{-4}	$[1.71 \times 10^{-4}, 0.00103]$	0.2137
model_type: tropical - vanilla head=pointer fishr_lambda=100	6.000×10^{-4}	$[-6.42 \times 10^{-5}, 0.00126]$	0.6618
model_type: tropical - vanilla head=pointer fishr_lambda=1000	6.000×10^{-4}	$[-6.42 \times 10^{-5}, 0.00126]$	0.6618
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	3.667×10^{-4}	$[-3.85 \times 10^{-4}, 0.00112]$	0.9887

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-1.333×10^{-4}	$[-5.04 \times 10^{-4}, 2.37 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-6.667×10^{-5}	$[-1.80 \times 10^{-4}, 4.67 \times 10^{-5}]$	0.8890
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	3.333×10^{-5}	$[-4.82 \times 10^{-4}, 5.49 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	1.667×10^{-4}	$[-8.68 \times 10^{-5}, 4.20 \times 10^{-4}]$	0.9935
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-6.667×10^{-5}	$[-4.96 \times 10^{-4}, 3.62 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	1.667×10^{-4}	$[-3.19 \times 10^{-4}, 6.52 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	2.000×10^{-4}	$[2.69 \times 10^{-5}, 3.73 \times 10^{-4}]$	0.2941
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-6.667×10^{-5}	$[-6.45 \times 10^{-4}, 5.11 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	1.333×10^{-4}	$[-3.98 \times 10^{-5}, 3.06 \times 10^{-4}]$	0.7944
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	1.667×10^{-4}	$[-8.68 \times 10^{-5}, 4.20 \times 10^{-4}]$	0.9935
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-6.667×10^{-4}	$[-9.59 \times 10^{-4}, -3.74 \times 10^{-4}]$	0.0702
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-8.000×10^{-4}	$[-0.00125, -3.51 \times 10^{-4}]$	0.1241
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-9.667×10^{-4}	$[-0.00142, -5.18 \times 10^{-4}]$	0.0746
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-8.000×10^{-4}	$[-0.00114, -4.60 \times 10^{-4}]$	0.0653

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-7.667×10^{-4}	[-0.00108, -4.53×10^{-4}]	0.0592
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.001033	[-0.00181, -2.54×10^{-4}]	0.1760
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-7.667×10^{-4}	[-0.00117, -3.63×10^{-4}]	0.1051
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-6.667×10^{-4}	[-0.0013, -2.88×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-1.333×10^{-4}	[-3.06×10^{-4}, 3.98×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-2.667×10^{-4}	[-5.03×10^{-4}, -3.07×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-4.000×10^{-4}	[-7.46×10^{-4}, -5.37×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-3.333×10^{-4}	[-6.26×10^{-4}, -4.07×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-3.333×10^{-4}	[-7.20×10^{-4}, 5.38×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-3.667×10^{-4}	[-6.36×10^{-4}, -9.68×10^{-5}]	0.1760
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-3.000×10^{-4}	[-5.70×10^{-4}, -3.02×10^{-5}]	0.2287
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-5.000×10^{-4}	[-8.27×10^{-4}, -1.73×10^{-4}]	0.1324
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-7.000×10^{-4}	[-0.0011, -3.02×10^{-4}]	0.1241
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-9.667×10^{-4}	[-0.00156, -3.74×10^{-4}]	0.1241
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-9.000×10^{-4}	[-0.00144, -3.60×10^{-4}]	0.1241
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-9.667×10^{-4}	[-0.00152, -4.11×10^{-4}]	0.1241
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-9.667×10^{-4}	[-0.00139, -5.43×10^{-4}]	0.0702
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-9.667×10^{-4}	[-0.00146, -4.73×10^{-4}]	0.1003
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-9.000×10^{-4}	[-0.0013, -4.97×10^{-4}]	0.0702
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-8.667×10^{-4}	[-0.00138, -3.51×10^{-4}]	0.1241

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.01748	[0.00665, 0.0286]	0.0134	5	0
adaptive	pointer	0.01	0.01623	[0.00705, 0.0233]	0.0104	5	0
adaptive	pointer	0.1	0.01326	[0.00542, 0.0211]	0.0103	5	0
adaptive	pointer	1	0.0203	[0.01, 0.027]	0.0116	5	0
adaptive	pointer	10	0.02095	[0.0101, 0.0285]	0.0122	5	0
adaptive	pointer	100	0.01778	[0.00866, 0.0242]	0.0104	5	0
adaptive	pointer	1000	0.01772	[0.00882, 0.0237]	0.0102	5	0
adaptive	pointer	1e+04	0.02054	[0.0103, 0.0274]	0.0118	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	7.477×10^{-4}	[0, 0.00224]	0.00167	5	0
adaptive	seq2seq	0.01	0	[0, 0]	0	5	0
adaptive	seq2seq	0.1	0	[0, 0]	0	5	0
adaptive	seq2seq	1	0	[0, 0]	0	5	0
adaptive	seq2seq	10	0	[0, 0]	0	5	0
adaptive	seq2seq	100	0	[0, 0]	0	5	0
adaptive	seq2seq	1000	0	[0, 0]	0	5	0
adaptive	seq2seq	1e+04	0	[0, 0]	0	5	0
tropical	pointer	0	0.004575	[0, 0.0137]	0.0102	5	0
tropical	pointer	0.01	0.003848	$[5.57 \times 10^{-4}$, 0.00758]	0.00425	5	0
tropical	pointer	0.1	0.005947	[0.00158, 0.0109]	0.00613	5	0
tropical	pointer	1	0.01099	$[4.71 \times 10^{-4}$, 0.0219]	0.0141	5	0
tropical	pointer	10	0.005823	$[5.19 \times 10^{-4}$, 0.0121]	0.00739	5	0
tropical	pointer	100	0.008088	[0.00105, 0.0151]	0.00897	5	0
tropical	pointer	1000	0.008224	$[4.91 \times 10^{-4}$, 0.0177]	0.0118	5	0
tropical	pointer	1e+04	0.003867	[0.00108, 0.00665]	0.00362	5	0
tropical	seq2seq	0	0	[0, 0]	0	5	0
tropical	seq2seq	0.01	0.003322	[0, 0.00997]	0.00743	5	0
tropical	seq2seq	0.1	0	[0, 0]	0	5	0
tropical	seq2seq	1	0	[0, 0]	0	5	0
tropical	seq2seq	10	0	[0, 0]	0	5	0
tropical	seq2seq	100	0.01102	[0.00268, 0.0194]	0.0116	5	0
tropical	seq2seq	1000	0.005198	[0, 0.0147]	0.0104	5	0
tropical	seq2seq	1e+04	0.003642	$[4.24 \times 10^{-4}$, 0.00865]	0.00568	5	0
vanilla	pointer	0	0.01302	[0.00425, 0.0218]	0.0111	5	0
vanilla	pointer	0.01	0.008373	$[4.86 \times 10^{-4}$, 0.018]	0.0111	5	0
vanilla	pointer	0.1	0.01023	[0.00354, 0.0186]	0.00962	5	0
vanilla	pointer	1	0.009962	[0.00266, 0.0183]	0.01	5	0
vanilla	pointer	10	0.009972	[0.00263, 0.0179]	0.0101	5	0
vanilla	pointer	100	0.00996	[0.00246, 0.0188]	0.0104	5	0
vanilla	pointer	1000	0.00977	[0.00183, 0.0182]	0.0107	5	0
vanilla	pointer	1e+04	0.009312	[0.00153, 0.0179]	0.0106	5	0
vanilla	seq2seq	0	0	[0, 0]	0	5	0
vanilla	seq2seq	0.01	0	[0, 0]	0	5	0
vanilla	seq2seq	0.1	0	[0, 0]	0	5	0
vanilla	seq2seq	1	0	[0, 0]	0	5	0
vanilla	seq2seq	10	0	[0, 0]	0	5	0
vanilla	seq2seq	100	0.001058	[0, 0.00317]	0.00237	5	0
vanilla	seq2seq	1000	0	[0, 0]	0	5	0
vanilla	seq2seq	1e+04	0	[0, 0]	0	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	1.438×10^{-5}	[-0.00882, 0.00479]	1.0000
adaptive	pointer	0.1	-0.001521	[-0.0136, 0.00342]	1.0000
adaptive	pointer	1	7.829×10^{-4}	[-0.00521, 0.0111]	1.0000
adaptive	pointer	10	4.566×10^{-4}	[-0.00474, 0.0125]	1.0000
adaptive	pointer	100	4.635×10^{-5}	[-0.0063, 0.00602]	1.0000
adaptive	pointer	1000	0	[-0.00665, 0.006]	1.0000
adaptive	pointer	1e+04	1.553×10^{-4}	[-0.00473, 0.0117]	1.0000
adaptive	seq2seq	0.01	0	[-0.00224, 0]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	0.1	0	[-0.00224, 0]	1.0000
adaptive	seq2seq	1	0	[-0.00224, 0]	1.0000
adaptive	seq2seq	10	0	[-0.00224, 0]	1.0000
adaptive	seq2seq	100	0	[-0.00224, 0]	1.0000
adaptive	seq2seq	1000	0	[-0.00224, 0]	1.0000
adaptive	seq2seq	1e+04	0	[-0.00224, 0]	1.0000
tropical	pointer	0.01	0	[-0.00748, 0.00419]	1.0000
tropical	pointer	0.1	0.002561	[-0.00374, 0.006]	1.0000
tropical	pointer	1	0.001329	$[2.66 \times 10^{-4}, 0.0175]$	1.0000
tropical	pointer	10	0	[-0.00755, 0.0103]	1.0000
tropical	pointer	100	0	[-0.00175, 0.01]	1.0000
tropical	pointer	1000	0.002457	$[4.91 \times 10^{-4}, 0.00751]$	1.0000
tropical	pointer	1e+04	0	[-0.0089, 0.00569]	1.0000
tropical	seq2seq	0.01	0	[0, 0.00997]	1.0000
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0.01338	[0.00268, 0.0194]	1.0000
tropical	seq2seq	1000	0	[0, 0.0147]	1.0000
tropical	seq2seq	1e+04	0.002121	$[4.24 \times 10^{-4}, 0.00865]$	1.0000
vanilla	pointer	0.01	-0.001074	[-0.0139, 0.00113]	1.0000
vanilla	pointer	0.1	-4.475×10^{-6}	[-0.00958, 0.00135]	1.0000
vanilla	pointer	1	-0.001341	$[-0.0089, 9.78 \times 10^{-4}]$	1.0000
vanilla	pointer	10	-0.001251	[-0.00896, 0.00103]	1.0000
vanilla	pointer	100	-0.001327	[-0.00932, 0.00135]	1.0000
vanilla	pointer	1000	-0.001174	$[-0.00837, 2.82 \times 10^{-4}]$	1.0000
vanilla	pointer	1e+04	-0.001678	$[-0.00933, 7.23 \times 10^{-5}]$	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0.00317]	1.0000
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.1505	[-0.0855, 0.411]	0.3881	0.11	0.4037
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	-0.02251	[-0.367, 0.36]	0.8979	-0.019	0.8872
tropical	seq2seq	0.382	[0.177, 0.613]	0.0235	0.219	0.0324
vanilla	pointer	-0.02425	[-0.386, 0.282]	0.8900	-0.0248	0.8476
vanilla	seq2seq	0.08575	[0.0857, 0.153]	0.6243	0.019	0.8516

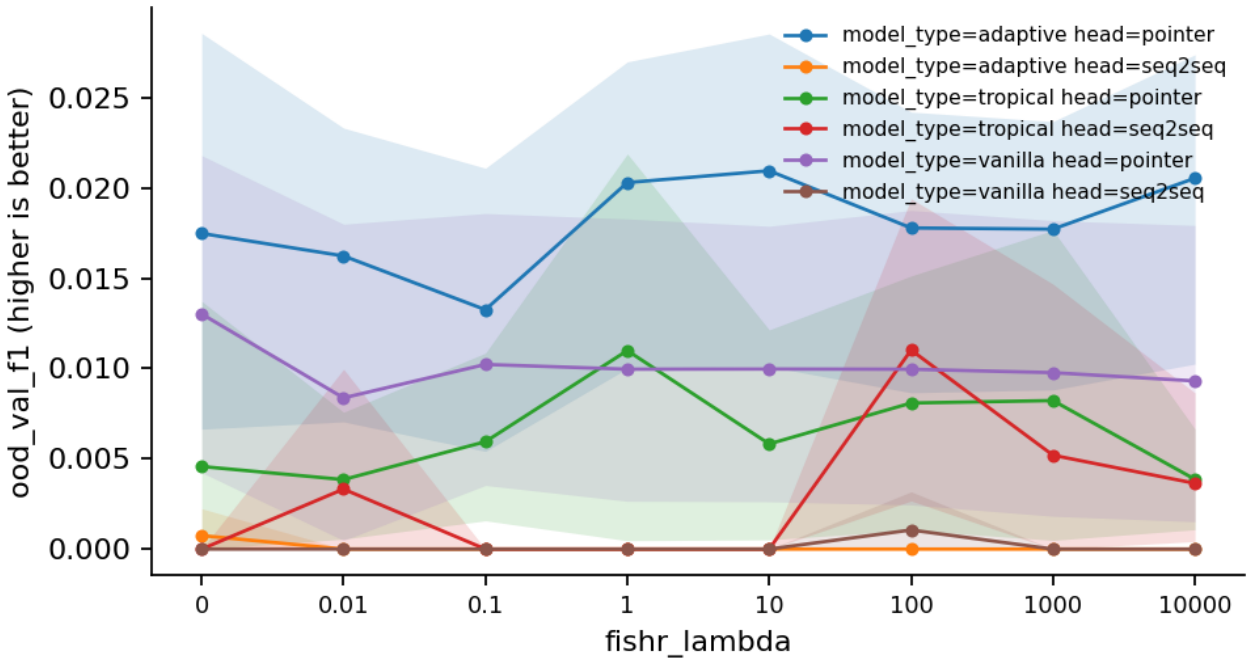


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.008446	[-0.0261, 0.00923]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.004525	[-0.0198, 0.0108]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.004279	[-0.0182, 0.00968]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.001031	[-0.023, 0.0251]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.004149	[-0.0231, 0.0148]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.001871	[-0.0102, 0.00643]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.001546	[-0.0152, 0.0121]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.005445	[-0.0163, 0.00541]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.003322	[-0.0059, 0.0125]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.009963	[-0.00426, 0.0242]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.005198	[-0.0077, 0.0181]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.003642	[-0.00341, 0.0107]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00446	[-0.015, 0.024]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.007861	[-0.00246, 0.0182]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.003031	[-0.00717, 0.0132]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01033	[-0.00246, 0.0231]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01098	[-0.00239, 0.0243]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.007824	[-0.00231, 0.018]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.007948	[-0.00303, 0.0189]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01123	[-0.00246, 0.0249]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	7.477×10^{-4}	[-0.00133, 0.00282]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.001058	[-0.004, 0.00188]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01673	[-0.00147, 0.0349]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01623	[0.00327, 0.0292]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01326	[5.09×10^{-4}, 0.026]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0203	[0.00588, 0.0347]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02095	[0.00579, 0.0361]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.01778	[0.00488, 0.0307]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01772	[0.00505, 0.0304]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02054	[0.00584, 0.0352]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.004575	[-0.00813, 0.0173]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	5.264×10^{-4}	[-0.0122, 0.0133]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.005947	[-0.00166, 0.0136]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01099	[-0.0065, 0.0285]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.005823	[-0.00335, 0.015]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.002933	[-0.0278, 0.0219]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.003026	[-0.0203, 0.0263]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	2.249×10^{-4}	[-0.00745, 0.0079]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01302	$[-7.71 \times 10^{-4}, 0.0268]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.008373	[-0.00543, 0.0222]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01023	[-0.00172, 0.0222]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.009962	[-0.00247, 0.0224]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.009972	[-0.00253, 0.0225]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.008901	[-0.00584, 0.0236]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.00977	[-0.00356, 0.0231]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.009312	[-0.00388, 0.0225]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9007	[0.831, 0.956]	0.0827	5	0
adaptive	pointer	0.01	0.9033	[0.856, 0.946]	0.0561	5	0
adaptive	pointer	0.1	0.925	[0.872, 0.962]	0.0598	5	0
adaptive	pointer	1	0.8888	[0.845, 0.936]	0.0589	5	0
adaptive	pointer	10	0.902	[0.86, 0.944]	0.0583	5	0
adaptive	pointer	100	0.9073	[0.863, 0.95]	0.0566	5	0
adaptive	pointer	1000	0.9006	[0.856, 0.941]	0.0533	5	0
adaptive	pointer	1e+04	0.8968	[0.85, 0.937]	0.056	5	0
adaptive	seq2seq	0	0.9689	[0.955, 0.976]	0.0154	5	0
adaptive	seq2seq	0.01	0.9762	[0.975, 0.977]	0.00141	5	0
adaptive	seq2seq	0.1	0.9759	[0.975, 0.976]	0.00111	5	0
adaptive	seq2seq	1	0.9762	[0.975, 0.977]	0.00141	5	0
adaptive	seq2seq	10	0.9762	[0.975, 0.977]	0.00141	5	0
adaptive	seq2seq	100	0.9762	[0.975, 0.977]	0.00141	5	0
adaptive	seq2seq	1000	0.9762	[0.975, 0.977]	0.00141	5	0
adaptive	seq2seq	1e+04	0.9761	[0.975, 0.977]	0.00132	5	0
tropical	pointer	0	0.9629	[0.939, 0.976]	0.0265	5	0
tropical	pointer	0.01	0.966	[0.955, 0.975]	0.0126	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.96	[0.939, 0.974]	0.0234	5	0
tropical	pointer	1	0.9457	[0.908, 0.973]	0.0418	5	0
tropical	pointer	10	0.9634	[0.952, 0.973]	0.0132	5	0
tropical	pointer	100	0.9679	[0.961, 0.974]	0.00825	5	0
tropical	pointer	1000	0.9479	[0.909, 0.972]	0.0436	5	0
tropical	pointer	1e+04	0.9657	[0.954, 0.976]	0.0139	5	0
tropical	seq2seq	0	0.9762	[0.975, 0.977]	0.00141	5	0
tropical	seq2seq	0.01	0.9636	[0.939, 0.977]	0.0277	5	0
tropical	seq2seq	0.1	0.9762	[0.975, 0.977]	0.00141	5	0
tropical	seq2seq	1	0.9756	[0.975, 0.976]	0.00104	5	0
tropical	seq2seq	10	0.9761	[0.975, 0.977]	0.00138	5	0
tropical	seq2seq	100	0.9449	[0.91, 0.972]	0.0402	5	0
tropical	seq2seq	1000	0.9474	[0.898, 0.976]	0.0546	5	0
tropical	seq2seq	1e+04	0.965	[0.956, 0.974]	0.0116	5	0
vanilla	pointer	0	0.8764	[0.793, 0.952]	0.109	5	0
vanilla	pointer	0.01	0.8878	[0.77, 0.969]	0.131	5	0
vanilla	pointer	0.1	0.8868	[0.773, 0.967]	0.127	5	0
vanilla	pointer	1	0.8811	[0.771, 0.963]	0.126	5	0
vanilla	pointer	10	0.8822	[0.779, 0.963]	0.122	5	0
vanilla	pointer	100	0.8833	[0.771, 0.965]	0.126	5	0
vanilla	pointer	1000	0.8841	[0.772, 0.966]	0.126	5	0
vanilla	pointer	1e+04	0.8835	[0.769, 0.966]	0.128	5	0
vanilla	seq2seq	0	0.9752	[0.974, 0.976]	0.00163	5	0
vanilla	seq2seq	0.01	0.9762	[0.975, 0.977]	0.00141	5	0
vanilla	seq2seq	0.1	0.9761	[0.975, 0.977]	0.00141	5	0
vanilla	seq2seq	1	0.9762	[0.975, 0.977]	0.00141	5	0
vanilla	seq2seq	10	0.9762	[0.975, 0.977]	0.00141	5	0
vanilla	seq2seq	100	0.9746	[0.972, 0.976]	0.00283	5	0
vanilla	seq2seq	1000	0.9733	[0.968, 0.976]	0.00563	5	0
vanilla	seq2seq	1e+04	0.9761	[0.975, 0.977]	0.00135	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.0026	[-0.104, 0.109]	1.0000
adaptive	pointer	0.1	0.02426	[-0.082, 0.131]	0.9828
adaptive	pointer	1	-0.0119	[-0.118, 0.0944]	0.9998
adaptive	pointer	10	0.00132	[-0.105, 0.108]	1.0000
adaptive	pointer	100	0.00658	[-0.0997, 0.113]	1.0000
adaptive	pointer	1000	-6.000×10^{-5}	[-0.106, 0.106]	1.0000
adaptive	pointer	1e+04	-0.0039	[-0.11, 0.102]	1.0000
adaptive	seq2seq	0.01	0.00728	[-0.00249, 0.017]	0.2135
adaptive	seq2seq	0.1	0.007	[-0.00277, 0.0168]	0.2459
adaptive	seq2seq	1	0.00728	[-0.00249, 0.017]	0.2135
adaptive	seq2seq	10	0.00728	[-0.00249, 0.017]	0.2135
adaptive	seq2seq	100	0.00728	[-0.00249, 0.017]	0.2136
adaptive	seq2seq	1000	0.00728	[-0.00249, 0.017]	0.2134
adaptive	seq2seq	1e+04	0.00718	[-0.00259, 0.0169]	0.2246
tropical	pointer	0.01	0.00314	[-0.0427, 0.0489]	1.0000
tropical	pointer	0.1	-0.00288	[-0.0487, 0.0429]	1.0000
tropical	pointer	1	-0.01716	[-0.063, 0.0286]	0.8347
tropical	pointer	10	5.200×10^{-4}	[-0.0453, 0.0463]	1.0000
tropical	pointer	100	0.00504	[-0.0408, 0.0508]	0.9998

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	-0.015	[-0.0608, 0.0308]	0.9024
tropical	pointer	1e+04	0.00282	[-0.043, 0.0486]	1.0000
tropical	seq2seq	0.01	-0.01254	[-0.0584, 0.0333]	0.9567
tropical	seq2seq	0.1	0	[-0.0458, 0.0458]	1.0000
tropical	seq2seq	1	-5.400×10^{-4}	[-0.0464, 0.0453]	1.0000
tropical	seq2seq	10	-2.000×10^{-5}	[-0.0459, 0.0458]	1.0000
tropical	seq2seq	100	-0.03122	[-0.0771, 0.0146]	0.2917
tropical	seq2seq	1000	-0.02872	[-0.0746, 0.0171]	0.3720
tropical	seq2seq	1e+04	-0.01118	[-0.057, 0.0347]	0.9756
vanilla	pointer	0.01	0.01144	[-0.206, 0.229]	1.0000
vanilla	pointer	0.1	0.01036	[-0.207, 0.228]	1.0000
vanilla	pointer	1	0.00466	[-0.213, 0.222]	1.0000
vanilla	pointer	10	0.00584	[-0.212, 0.223]	1.0000
vanilla	pointer	100	0.0069	[-0.211, 0.225]	1.0000
vanilla	pointer	1000	0.00774	[-0.21, 0.225]	1.0000
vanilla	pointer	1e+04	0.00714	[-0.21, 0.225]	1.0000
vanilla	seq2seq	0.01	9.600×10^{-4}	[-0.0035, 0.00542]	0.9875
vanilla	seq2seq	0.1	9.200×10^{-4}	[-0.00354, 0.00538]	0.9901
vanilla	seq2seq	1	9.600×10^{-4}	[-0.0035, 0.00542]	0.9875
vanilla	seq2seq	10	9.600×10^{-4}	[-0.0035, 0.00542]	0.9875
vanilla	seq2seq	100	-6.200×10^{-4}	[-0.00508, 0.00384]	0.9991
vanilla	seq2seq	1000	-0.00192	[-0.00638, 0.00254]	0.7357
vanilla	seq2seq	1e+04	9.000×10^{-4}	[-0.00356, 0.00536]	0.9913

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.001774	[-0.00512, 5.16×10^{-4}]	0.6974	-0.0495	0.7197
adaptive	seq2seq	9.286×10^{-6}	[-8.57×10^{-6}, 3.86×10^{-5}]	0.9307	-0.0229	0.8528
tropical	pointer	-1.071×10^{-4}	[-0.00228, 0.00179]	0.9608	0.0133	0.9246
tropical	seq2seq	-0.003001	[-0.00561, -9.54×10^{-4}]	0.2156	-0.24	0.0688
vanilla	pointer	-5.679×10^{-4}	[-0.00167, 3.09×10^{-4}]	0.9544	-0.00952	0.9484
vanilla	seq2seq	-2.657×10^{-4}	[-7.99×10^{-4}, 6.43×10^{-6}]	0.2391	-0.118	0.3729

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0865	[-0.0512, 0.224]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0782	[-0.0847, 0.241]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.07326	[-0.0842, 0.231]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.06468	[-0.0963, 0.226]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.08118	[-0.0749, 0.237]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.08464	[-0.0673, 0.237]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.06376	[-0.09, 0.217]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.08218	[-0.0607, 0.225]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	9.600×10^{-4}	[-0.00171, 0.00363]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.01254	[-0.0469, 0.0219]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	4.000×10^{-5}	$[-7.11 \times 10^{-5}, 1.51 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-5.400×10^{-4}	[-0.00184, 7.55×10^{-4}]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-2.000×10^{-5}	$[-7.55 \times 10^{-5}, 3.55 \times 10^{-5}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.02964	[-0.081, 0.0217]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.02584	[-0.0952, 0.0436]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01112	[-0.0247, 0.00247]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0243	[-0.105, 0.153]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01546	[-0.0857, 0.117]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0382	[-0.0608, 0.137]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00774	[-0.0931, 0.109]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01978	[-0.0696, 0.109]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02398	[-0.0747, 0.123]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0165	[-0.0843, 0.117]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01326	[-0.089, 0.116]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00632	[-0.0239, 0.0112]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-2.400×10^{-4}	[-0.00105, 5.72×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00158	[-0.00281, 0.00597]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00288	[-0.00512, 0.0109]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-4.000×10^{-5}	$[-1.08 \times 10^{-4}, 2.80 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.06818	[-0.181, 0.0444]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.07286	[-0.142, -0.00374]	0.9021

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.05092	[-0.126, 0.0238]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.08736	[-0.159, -0.0153]	0.6766
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.07414	[-0.146, -0.00254]	0.9050
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.06888	[-0.138, 5.75×10^{-4}]	0.9727
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.07552	[-0.141, -0.0101]	0.7513
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.07926	[-0.148, -0.0107]	0.7513
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.01326	[-0.0446, 0.0181]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.00242	[-0.0409, 0.0458]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01614	[-0.0439, 0.0116]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.02988	[-0.0806, 0.0208]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01272	[-0.0286, 0.00314]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.023	[-0.0353, 0.0813]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	4.600×10^{-4}	[-0.0998, 0.101]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	7.400×10^{-4}	[-0.0235, 0.025]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.0988	[-0.235, 0.0373]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.08832	[-0.25, 0.0734]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.08936	[-0.247, 0.0681]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0951	[-0.251, 0.0604]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.09392	[-0.246, 0.0577]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.09128	[-0.25, 0.067]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.08914	[-0.249, 0.0709]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.09256	[-0.252, 0.0664]	1.0000

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.005681	[0.00107, 0.0131]	0.00836	5	0
adaptive	pointer	0.01	0.004628	[2.15×10^{-4}, 0.0116]	0.0076	5	0
adaptive	pointer	0.1	0.006896	[0.00128, 0.0137]	0.00806	5	0
adaptive	pointer	1	0.005111	[4.27×10^{-4}, 0.0124]	0.00807	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.005446	[9.55×10^{-4}, 0.0117]	0.00721	5	0
adaptive	pointer	100	0.007034	[0.00238, 0.0134]	0.00746	5	0
adaptive	pointer	1000	0.004529	[4.27×10^{-4}, 0.0118]	0.00808	5	0
adaptive	pointer	1e+04	0.005097	[0.00107, 0.0112]	0.00662	5	0
adaptive	seq2seq	0	0	[0, 0]	0	5	0
adaptive	seq2seq	0.01	0	[0, 0]	0	5	0
adaptive	seq2seq	0.1	0	[0, 0]	0	5	0
adaptive	seq2seq	1	2.247×10^{-4}	[0, 6.74×10^{-4}]	5.02×10^{-4}	5	0
adaptive	seq2seq	10	0	[0, 0]	0	5	0
adaptive	seq2seq	100	0	[0, 0]	0	5	0
adaptive	seq2seq	1000	0	[0, 0]	0	5	0
adaptive	seq2seq	1e+04	0.002666	[5.42×10^{-4}, 0.00484]	0.00285	5	0
tropical	pointer	0	7.762×10^{-4}	[0, 0.0019]	0.00122	5	0
tropical	pointer	0.01	0.00182	[0, 0.00501]	0.00347	5	0
tropical	pointer	0.1	3.261×10^{-4}	[0, 7.54×10^{-4}]	4.81×10^{-4}	5	0
tropical	pointer	1	1.109×10^{-4}	[0, 3.33×10^{-4}]	2.48×10^{-4}	5	0
tropical	pointer	10	5.168×10^{-4}	[0, 0.00155]	0.00116	5	0
tropical	pointer	100	0	[0, 0]	0	5	0
tropical	pointer	1000	5.438×10^{-4}	[0, 0.00141]	9.34×10^{-4}	5	0
tropical	pointer	1e+04	7.388×10^{-4}	[0, 0.00222]	0.00165	5	0
tropical	seq2seq	0	0	[0, 0]	0	5	0
tropical	seq2seq	0.01	6.476×10^{-4}	[0, 0.00194]	0.00145	5	0
tropical	seq2seq	0.1	3.223×10^{-4}	[0, 9.67×10^{-4}]	7.21×10^{-4}	5	0
tropical	seq2seq	1	0.007066	[0, 0.0212]	0.0158	5	0
tropical	seq2seq	10	0.003491	[0, 0.0105]	0.00781	5	0
tropical	seq2seq	100	0.001924	[0, 0.00577]	0.0043	5	0
tropical	seq2seq	1000	5.504×10^{-4}	[0, 0.00122]	7.83×10^{-4}	5	0
tropical	seq2seq	1e+04	0.002246	[1.12×10^{-4}, 0.00595]	0.00411	5	0
vanilla	pointer	0	0.019	[7.56×10^{-4}, 0.0536]	0.0387	5	0
vanilla	pointer	0.01	0.01393	[8.48×10^{-4}, 0.0372]	0.0257	5	0
vanilla	pointer	0.1	0.0138	[8.46×10^{-4}, 0.0368]	0.0255	5	0
vanilla	pointer	1	0.01416	[7.36×10^{-4}, 0.0371]	0.0253	5	0
vanilla	pointer	10	0.0145	[7.37×10^{-4}, 0.0379]	0.0258	5	0
vanilla	pointer	100	0.01423	[7.34×10^{-4}, 0.0379]	0.0262	5	0
vanilla	pointer	1000	0.01377	[4.13×10^{-4}, 0.0374]	0.026	5	0
vanilla	pointer	1e+04	0.01363	[9.62×10^{-4}, 0.0365]	0.0253	5	0
vanilla	seq2seq	0	0	[0, 0]	0	5	0
vanilla	seq2seq	0.01	0	[0, 0]	0	5	0
vanilla	seq2seq	0.1	0	[0, 0]	0	5	0
vanilla	seq2seq	1	0	[0, 0]	0	5	0
vanilla	seq2seq	10	0	[0, 0]	0	5	0
vanilla	seq2seq	100	0	[0, 0]	0	5	0
vanilla	seq2seq	1000	0	[0, 0]	0	5	0
vanilla	seq2seq	1e+04	0.001845	[5.42×10^{-4}, 0.00313]	0.0017	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.00256	[-0.00262, 0.001]	1.0000
adaptive	pointer	0.1	0	[-0.00132, 0.00406]	1.0000
adaptive	pointer	1	-0.001292	[-0.00215, 0.00138]	1.0000
adaptive	pointer	10	-3.779×10^{-5}	[-0.00213, 0.00196]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	100	0.002225	$[-8.85 \times 10^{-4}$, 0.00359]	1.0000
adaptive	pointer	1000	-0.001507	[-0.0022, -1.09×10^{-4}]	1.0000
adaptive	pointer	1e+04	-5.509×10^{-4}	[-0.00278, 0.00157]	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 6.74×10^{-4}]	1.0000
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0.002711	[5.42×10^{-4}, 0.00484]	0.5000
tropical	pointer	0.01	0	[-0.00101, 0.00414]	1.0000
tropical	pointer	0.1	0	[-0.00134, 0]	1.0000
tropical	pointer	1	0	[-0.00177, 2.22×10^{-4}]	1.0000
tropical	pointer	10	0	[-0.00168, 9.02×10^{-4}]	1.0000
tropical	pointer	100	0	[-0.00177, 0]	1.0000
tropical	pointer	1000	0	[-0.00157, 7.59×10^{-4}]	1.0000
tropical	pointer	1e+04	0	[-0.00168, 0.00157]	1.0000
tropical	seq2seq	0.01	0	[0, 0.00194]	1.0000
tropical	seq2seq	0.1	0	[0, 9.67×10^{-4}]	1.0000
tropical	seq2seq	1	0	[0, 0.0212]	1.0000
tropical	seq2seq	10	0	[0, 0.0105]	1.0000
tropical	seq2seq	100	0	[0, 0.00577]	1.0000
tropical	seq2seq	1000	0	[0, 0.0012]	1.0000
tropical	seq2seq	1e+04	5.599×10^{-4}	[1.12×10^{-4}, 0.00595]	1.0000
vanilla	pointer	0.01	0	[-0.017, 0.00224]	1.0000
vanilla	pointer	0.1	0	[-0.0174, 0.00224]	1.0000
vanilla	pointer	1	0	[-0.0175, 0.00343]	1.0000
vanilla	pointer	10	0	[-0.0168, 0.00375]	1.0000
vanilla	pointer	100	0	[-0.0164, 0.00256]	1.0000
vanilla	pointer	1000	-5.546×10^{-4}	[-0.017, 0.00215]	1.0000
vanilla	pointer	1e+04	0	[-0.0177, 0.00196]	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0.002709	[5.42×10^{-4}, 0.0031]	0.2500

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.04184	[-0.198, 0.256]	0.8114	0.0248	0.8658
adaptive	seq2seq	0.3737	[0.2, 0.549]	0.0270	0.158	0.0262
tropical	pointer	-0.1197	[-0.486, 0.171]	0.4935	-0.0762	0.4729
tropical	seq2seq	0.204	[-0.183, 0.667]	0.2398	0.12	0.2631
vanilla	pointer	-0.009249	[-0.0302, 0.0291]	0.9579	-0.00952	0.9462
vanilla	seq2seq	0.4587	[0.257, 0.611]	0.0056	0.171	0.0034

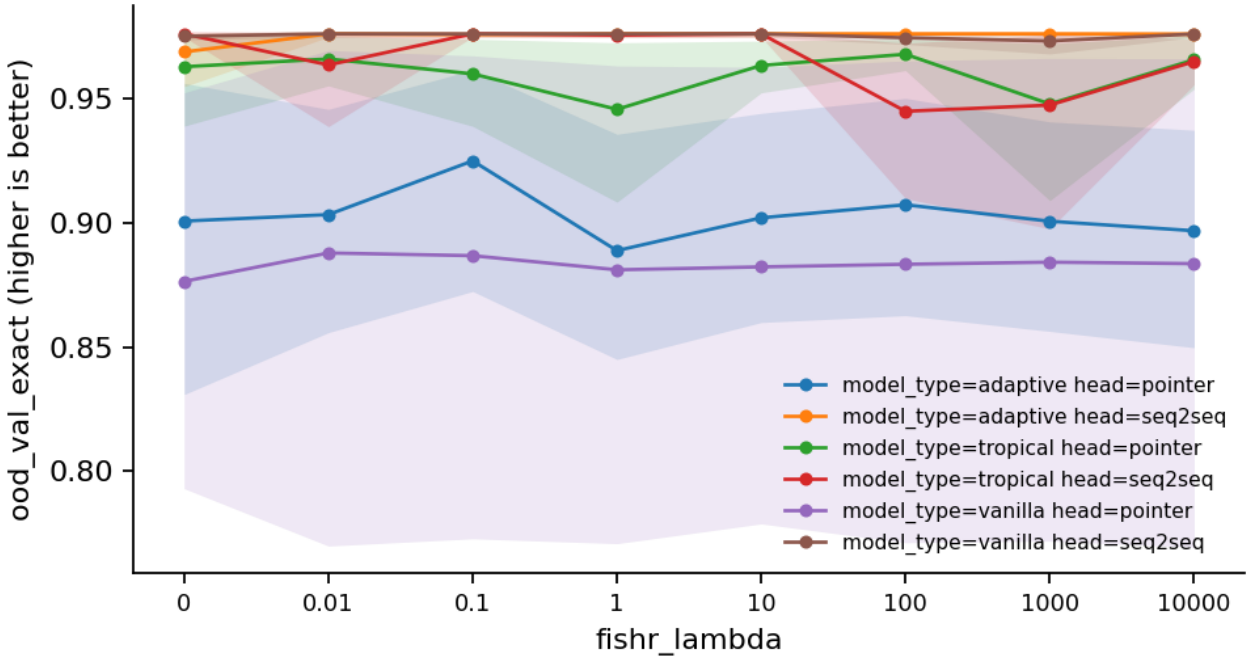


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

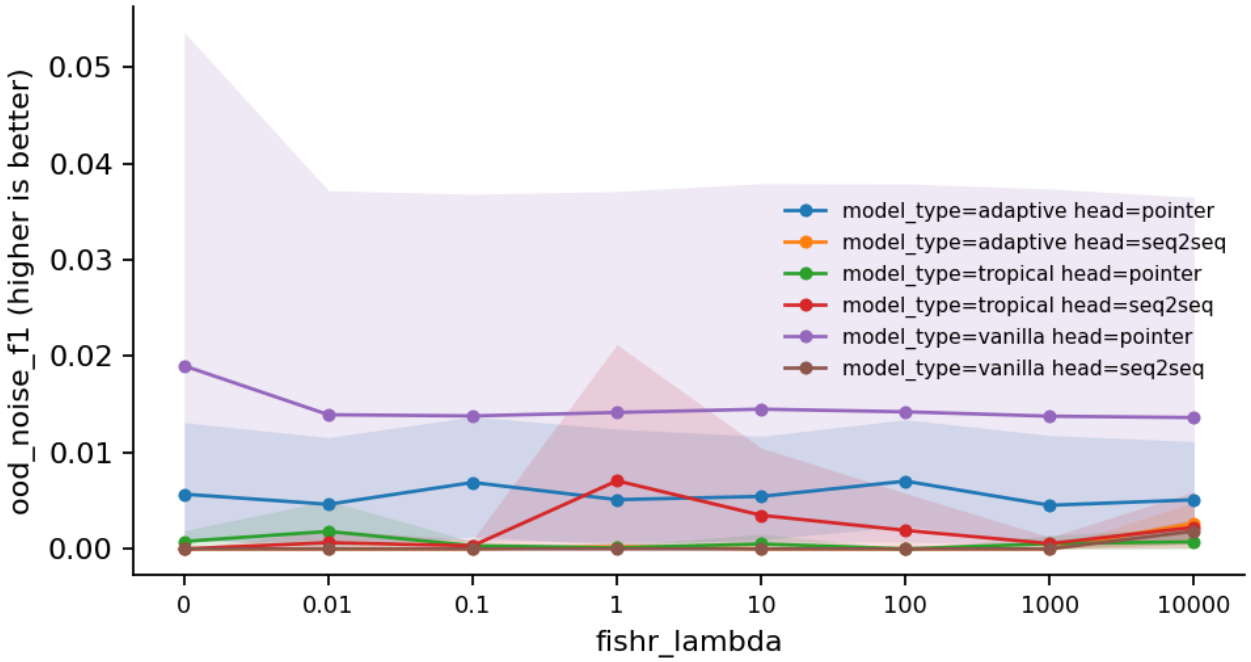


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.01823	[-0.0668, 0.0304]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.01211	[-0.0452, 0.021]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.01347	[-0.0453, 0.0183]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.01405	[-0.0451, 0.017]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.01399	[-0.0462, 0.0182]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.01423	[-0.0467, 0.0182]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.01323	[-0.0458, 0.0193]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.01289	[-0.0447, 0.0189]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	6.476×10^{-4}	[-0.00115, 0.00245]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	3.223×10^{-4}	$[-5.73 \times 10^{-4}, 0.00122]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.007066	[-0.0126, 0.0267]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.003491	[-0.0062, 0.0132]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.001924	[-0.00342, 0.00726]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	5.504×10^{-4}	$[-4.21 \times 10^{-4}, 0.00152]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	4.002×10^{-4}	[-0.00631, 0.00711]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.01332	[-0.0512, 0.0245]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.009297	[-0.0323, 0.0137]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.006905	[-0.0295, 0.0157]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.009049	[-0.031, 0.0129]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.009057	[-0.0326, 0.0145]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.007195	[-0.0307, 0.0163]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.009245	[-0.0316, 0.0132]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.008532	[-0.0321, 0.0151]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	2.247×10^{-4}	$[-3.99 \times 10^{-4}, 8.49 \times 10^{-4}]$	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	8.210×10^{-4}	[-0.00111, 0.00276]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.005681	[-0.00469, 0.0161]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.004628	[-0.00481, 0.0141]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.006896	[-0.00311, 0.0169]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.004886	[-0.00537, 0.0151]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.005446	[-0.00351, 0.0144]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.007034	[-0.00223, 0.0163]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.004529	[-0.00551, 0.0146]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.00243	[-0.0071, 0.012]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	7.762×10^{-4}	$[-7.44 \times 10^{-4}, 0.0023]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.001172	[-0.00397, 0.00631]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	3.740×10^{-6}	[-0.00125, 0.00125]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.006956	[-0.0267, 0.0127]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.002974	[-0.0131, 0.00717]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.001924	[-0.00726, 0.00342]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-6.569×10^{-6}	[-0.00186, 0.00185]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.001507	[-0.00458, 0.00156]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.019	[-0.029, 0.067]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01393	[-0.018, 0.0459]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0138	[-0.0178, 0.0454]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01416	[-0.0172, 0.0455]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0145	[-0.0175, 0.0465]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01423	[-0.0182, 0.0467]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01377	[-0.0185, 0.0461]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01178	[-0.0193, 0.0429]	1.0000

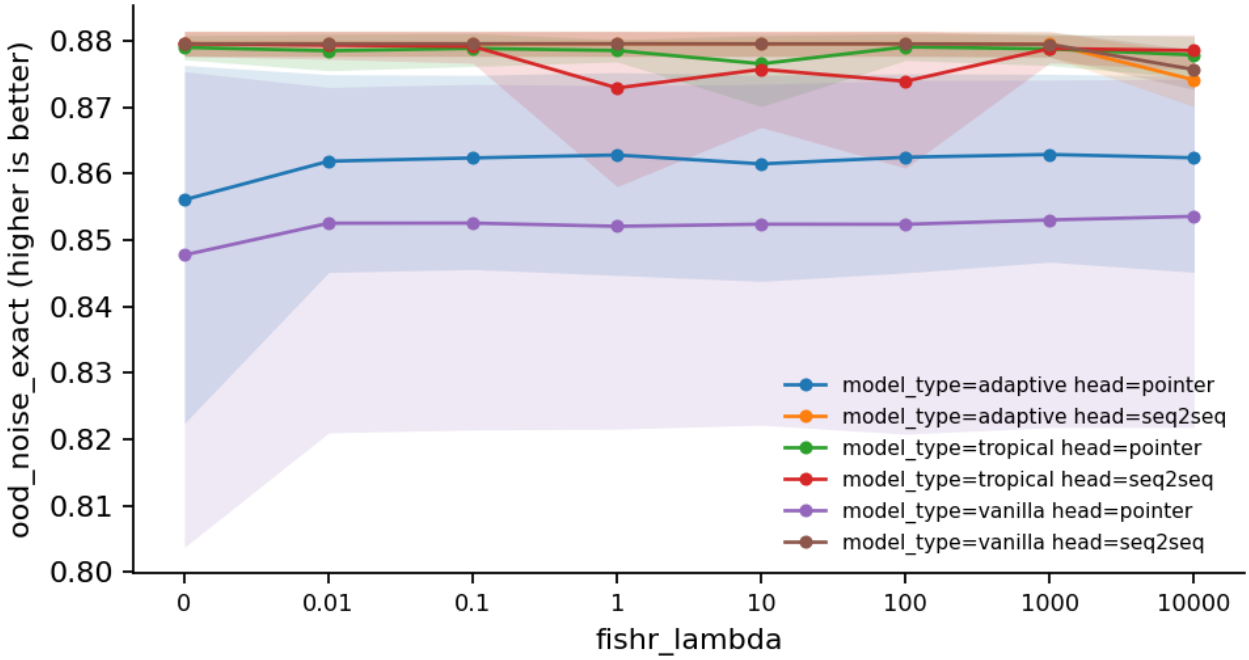


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8561	[0.822, 0.876]	0.0375	5	0
adaptive	pointer	0.01	0.8619	[0.845, 0.875]	0.0195	5	0
adaptive	pointer	0.1	0.8624	[0.846, 0.875]	0.0196	5	0
adaptive	pointer	1	0.8628	[0.845, 0.875]	0.0209	5	0
adaptive	pointer	10	0.8615	[0.844, 0.875]	0.0205	5	0
adaptive	pointer	100	0.8625	[0.845, 0.875]	0.02	5	0
adaptive	pointer	1000	0.8629	[0.847, 0.875]	0.0187	5	0
adaptive	pointer	1e+04	0.8624	[0.845, 0.875]	0.0188	5	0
adaptive	seq2seq	0	0.8796	[0.878, 0.881]	0.0025	5	0
adaptive	seq2seq	0.01	0.8796	[0.878, 0.881]	0.0025	5	0
adaptive	seq2seq	0.1	0.8796	[0.878, 0.881]	0.00248	5	0
adaptive	seq2seq	1	0.8796	[0.878, 0.881]	0.0025	5	0
adaptive	seq2seq	10	0.8796	[0.878, 0.881]	0.0025	5	0
adaptive	seq2seq	100	0.8796	[0.878, 0.881]	0.0025	5	0
adaptive	seq2seq	1000	0.8795	[0.878, 0.881]	0.00256	5	0
adaptive	seq2seq	1e+04	0.8741	[0.87, 0.878]	0.00549	5	0
tropical	pointer	0	0.879	[0.877, 0.881]	0.00235	5	0
tropical	pointer	0.01	0.8785	[0.876, 0.881]	0.00342	5	0
tropical	pointer	0.1	0.8789	[0.876, 0.881]	0.00317	5	0
tropical	pointer	1	0.8786	[0.877, 0.88]	0.00219	5	0
tropical	pointer	10	0.8766	[0.87, 0.881]	0.00713	5	0
tropical	pointer	100	0.8791	[0.877, 0.881]	0.0026	5	0
tropical	pointer	1000	0.8788	[0.876, 0.881]	0.00306	5	0
tropical	pointer	1e+04	0.8779	[0.874, 0.881]	0.00435	5	0
tropical	seq2seq	0	0.8796	[0.878, 0.881]	0.0025	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.8794	[0.877, 0.881]	0.0028	5	0
tropical	seq2seq	0.1	0.8792	[0.877, 0.881]	0.00309	5	0
tropical	seq2seq	1	0.8729	[0.858, 0.881]	0.0162	5	0
tropical	seq2seq	10	0.8757	[0.867, 0.881]	0.0103	5	0
tropical	seq2seq	100	0.8739	[0.861, 0.881]	0.0143	5	0
tropical	seq2seq	1000	0.8789	[0.877, 0.881]	0.00274	5	0
tropical	seq2seq	1e+04	0.8786	[0.876, 0.881]	0.00337	5	0
vanilla	pointer	0	0.8478	[0.804, 0.875]	0.0512	5	0
vanilla	pointer	0.01	0.8525	[0.821, 0.873]	0.0354	5	0
vanilla	pointer	0.1	0.8526	[0.821, 0.873]	0.0349	5	0
vanilla	pointer	1	0.8521	[0.822, 0.873]	0.0342	5	0
vanilla	pointer	10	0.8524	[0.822, 0.873]	0.0344	5	0
vanilla	pointer	100	0.8524	[0.821, 0.874]	0.0354	5	0
vanilla	pointer	1000	0.853	[0.822, 0.874]	0.035	5	0
vanilla	pointer	1e+04	0.8536	[0.822, 0.874]	0.0355	5	0
vanilla	seq2seq	0	0.8796	[0.878, 0.881]	0.0025	5	0
vanilla	seq2seq	0.01	0.8796	[0.878, 0.881]	0.0025	5	0
vanilla	seq2seq	0.1	0.8796	[0.878, 0.881]	0.0025	5	0
vanilla	seq2seq	1	0.8796	[0.878, 0.881]	0.0025	5	0
vanilla	seq2seq	10	0.8796	[0.878, 0.881]	0.00248	5	0
vanilla	seq2seq	100	0.8796	[0.878, 0.881]	0.00248	5	0
vanilla	seq2seq	1000	0.8795	[0.878, 0.881]	0.00256	5	0
vanilla	seq2seq	1e+04	0.8757	[0.873, 0.879]	0.00396	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.0058	[-0.0339, 0.0455]	0.9987	
adaptive	pointer	0.1	0.0063	[-0.0334, 0.046]	0.9979	
adaptive	pointer	1	0.00674	[-0.033, 0.0464]	0.9968	
adaptive	pointer	10	0.0054	[-0.0343, 0.0451]	0.9992	
adaptive	pointer	100	0.0064	[-0.0333, 0.0461]	0.9977	
adaptive	pointer	1000	0.00682	[-0.0329, 0.0465]	0.9966	
adaptive	pointer	1e+04	0.00632	[-0.0334, 0.046]	0.9979	
adaptive	seq2seq	0.01	0	[-0.00532, 0.00532]	1.0000	
adaptive	seq2seq	0.1	-2.000×10^{-5}	[-0.00534, 0.0053]	1.0000	
adaptive	seq2seq	1	0	[-0.00532, 0.00532]	1.0000	
adaptive	seq2seq	10	0	[-0.00532, 0.00532]	1.0000	
adaptive	seq2seq	100	0	[-0.00532, 0.00532]	1.0000	
adaptive	seq2seq	1000	-4.000×10^{-5}	[-0.00536, 0.00528]	1.0000	
adaptive	seq2seq	1e+04	-0.00546	[-0.0108, -1.43×10^{-4}]	0.0423	*
tropical	pointer	0.01	-5.000×10^{-4}	[-0.00721, 0.00621]	1.0000	
tropical	pointer	0.1	-1.400×10^{-4}	[-0.00685, 0.00657]	1.0000	
tropical	pointer	1	-4.600×10^{-4}	[-0.00717, 0.00625]	1.0000	
tropical	pointer	10	-0.00246	[-0.00917, 0.00425]	0.8473	
tropical	pointer	100	8.000×10^{-5}	[-0.00663, 0.00679]	1.0000	
tropical	pointer	1000	-2.200×10^{-4}	[-0.00693, 0.00649]	1.0000	
tropical	pointer	1e+04	-0.00114	[-0.00785, 0.00557]	0.9968	
tropical	seq2seq	0.01	-2.200×10^{-4}	[-0.0155, 0.0151]	1.0000	
tropical	seq2seq	0.1	-4.000×10^{-4}	[-0.0157, 0.0149]	1.0000	
tropical	seq2seq	1	-0.00666	[-0.022, 0.00865]	0.7279	
tropical	seq2seq	10	-0.00386	[-0.0192, 0.0115]	0.9711	
tropical	seq2seq	100	-0.00566	[-0.021, 0.00965]	0.8427	

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1000	-7.200×10^{-4}	[-0.016, 0.0146]	1.0000
tropical	seq2seq	1e+04	-0.00102	[-0.0163, 0.0143]	1.0000
vanilla	pointer	0.01	0.00478	[-0.0605, 0.0701]	1.0000
vanilla	pointer	0.1	0.0048	[-0.0605, 0.0701]	1.0000
vanilla	pointer	1	0.00432	[-0.061, 0.0696]	1.0000
vanilla	pointer	10	0.00464	[-0.0607, 0.0699]	1.0000
vanilla	pointer	100	0.00462	[-0.0607, 0.0699]	1.0000
vanilla	pointer	1000	0.00528	[-0.06, 0.0706]	1.0000
vanilla	pointer	1e+04	0.00582	[-0.0595, 0.0711]	0.9999
vanilla	seq2seq	0.01	0	[-0.00477, 0.00477]	1.0000
vanilla	seq2seq	0.1	0	[-0.00477, 0.00477]	1.0000
vanilla	seq2seq	1	0	[-0.00477, 0.00477]	1.0000
vanilla	seq2seq	10	-2.000×10^{-5}	[-0.00479, 0.00475]	1.0000
vanilla	seq2seq	100	-2.000×10^{-5}	[-0.00479, 0.00475]	1.0000
vanilla	seq2seq	1000	-8.000×10^{-5}	[-0.00485, 0.00469]	1.0000
vanilla	seq2seq	1e+04	-0.0039	[-0.00867, 8.67×10^{-4}]	0.1458

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	8.071×10^{-5}	$[-5.14 \times 10^{-5}, 2.57 \times 10^{-4}]$	0.9584	-0.00762	0.9666
adaptive	seq2seq	-5.864×10^{-4}	$[-9.98 \times 10^{-4}, -1.75 \times 10^{-4}]$	0.0408	-0.261	0.0492
tropical	pointer	-5.500×10^{-5}	$[-1.24 \times 10^{-4}, 1.43 \times 10^{-5}]$	0.8645	-0.0381	0.7734
tropical	seq2seq	-7.286×10^{-5}	$[-2.21 \times 10^{-4}, 8.14 \times 10^{-5}]$	0.9240	-0.0838	0.5229
vanilla	pointer	1.564×10^{-4}	$[4.29 \times 10^{-6}, 3.37 \times 10^{-4}]$	0.9545	0.0495	0.7209
vanilla	seq2seq	-4.243×10^{-4}	$[-7.35 \times 10^{-4}, -1.48 \times 10^{-4}]$	0.0787	-0.227	0.0892

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.03126	[-0.0331, 0.0957]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.02598	[-0.0194, 0.0714]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.02632	[-0.0187, 0.0714]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.02648	[-0.0155, 0.0684]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02416	[-0.0216, 0.0699]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.02672	[-0.0185, 0.0719]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.02576	[-0.0194, 0.0709]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0243	[-0.0215, 0.0701]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-2.200×10^{-4}	$[-5.97 \times 10^{-4}, 1.57 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-4.000×10^{-4}	$[-0.00138, 5.78 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.00666	[-0.0247, 0.0113]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.00384	[-0.0146, 0.00689]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.00564	[-0.0214, 0.0101]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-6.400×10^{-4}	$[-0.00183, 5.46 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.00288	[-0.00288, 0.00864]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00832	[-0.0109, 0.0275]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00934	[-0.0111, 0.0297]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00982	[-0.00943, 0.0291]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01074	[-0.00601, 0.0275]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00908	[-0.00857, 0.0267]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0101	[-0.00929, 0.0295]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00986	[-0.0108, 0.0305]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00882	[-0.0124, 0.03]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-2.000×10^{-5}	$[-7.55 \times 10^{-5}, 3.55 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	2.000×10^{-5}	$[-3.55 \times 10^{-5}, 7.55 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	2.000×10^{-5}	$[-3.55 \times 10^{-5}, 7.55 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	4.000×10^{-5}	$[-2.80 \times 10^{-5}, 1.08 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00156	[-0.005, 0.00188]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.0235	[-0.071, 0.024]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0177	[-0.0434, 0.00801]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01718	[-0.0429, 0.0085]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01676	[-0.0439, 0.0104]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0181	[-0.0448, 0.00864]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.0171	[-0.0433, 0.00908]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01664	[-0.0413, 0.00803]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.01172	[-0.036, 0.0126]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-5.600×10^{-4}	[-0.00117, 5.21×10^{-5}]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-8.400×10^{-4}	[-0.0024, 7.25×10^{-4}]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-3.000×10^{-4}	[-0.00206, 0.00146]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.00564	[-0.013, 0.0243]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	8.400×10^{-4}	[-0.0131, 0.0148]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.00518	[-0.0105, 0.0208]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-6.000×10^{-5}	[-0.00143, 0.00131]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-6.800×10^{-4}	[-0.00237, 0.00101]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.03182	[-0.0967, 0.0331]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.02704	[-0.0724, 0.0183]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02702	[-0.0717, 0.0177]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0275	[-0.0712, 0.0162]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02716	[-0.0711, 0.0168]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.02718	[-0.0725, 0.0182]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.02646	[-0.0713, 0.0184]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.0221	[-0.0642, 0.02]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.0183	[0.011, 0.0253]	0.00919	5	0
adaptive	pointer	0.01	0.01853	[0.0132, 0.0247]	0.00772	5	0
adaptive	pointer	0.1	0.01646	[0.013, 0.0189]	0.00381	5	0
adaptive	pointer	1	0.02208	[0.014, 0.0324]	0.012	5	0
adaptive	pointer	10	0.02032	[0.0131, 0.0297]	0.0111	5	0
adaptive	pointer	100	0.01704	[0.0135, 0.0203]	0.0044	5	0
adaptive	pointer	1000	0.01873	[0.0131, 0.0262]	0.00879	5	0
adaptive	pointer	1e+04	0.01984	[0.0131, 0.0275]	0.00959	5	0
adaptive	seq2seq	0	2.201×10^{-4}	[0, 6.60×10^{-4}]	4.92×10^{-4}	5	0
adaptive	seq2seq	0.01	0	[0, 0]	0	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	1.102×10^{-4}	[0, 3.31×10^{-4}]	2.46×10^{-4}	5	0
adaptive	seq2seq	1	0	[0, 0]	0	5	0
adaptive	seq2seq	10	3.304×10^{-4}	[0, 9.91×10^{-4}]	7.39×10^{-4}	5	0
adaptive	seq2seq	100	4.404×10^{-4}	[0, 0.00132]	9.85×10^{-4}	5	0
adaptive	seq2seq	1000	0.001826	[8.42×10^{-4}, 0.0028]	0.0013	5	0
adaptive	seq2seq	1e+04	0.004786	[0.0031, 0.00643]	0.00215	5	0
tropical	pointer	0	0.003455	[0.00161, 0.00544]	0.00245	5	0
tropical	pointer	0.01	0.00496	[0.00223, 0.00792]	0.00355	5	0
tropical	pointer	0.1	0.006651	[0.00451, 0.00848]	0.00258	5	0
tropical	pointer	1	0.006365	[0.00171, 0.011]	0.00588	5	0
tropical	pointer	10	0.005494	[0.00223, 0.00942]	0.00453	5	0
tropical	pointer	100	0.005284	[0.00195, 0.00865]	0.00441	5	0
tropical	pointer	1000	0.005935	[0.00224, 0.00945]	0.00468	5	0
tropical	pointer	1e+04	0.003875	[0.00277, 0.00488]	0.00138	5	0
tropical	seq2seq	0	0	[0, 0]	0	5	0
tropical	seq2seq	0.01	0.004865	[0, 0.0146]	0.0109	5	0
tropical	seq2seq	0.1	6.327×10^{-4}	[0, 0.0019]	0.00141	5	0
tropical	seq2seq	1	0.003993	[0, 0.0103]	0.00678	5	0
tropical	seq2seq	10	0.003768	[0, 0.0109]	0.00784	5	0
tropical	seq2seq	100	0	[0, 0]	0	5	0
tropical	seq2seq	1000	3.167×10^{-4}	[0, 9.50×10^{-4}]	7.08×10^{-4}	5	0
tropical	seq2seq	1e+04	0.002674	[0.00129, 0.00405]	0.00189	5	0
vanilla	pointer	0	0.0196	[0.0159, 0.0221]	0.00411	5	0
vanilla	pointer	0.01	0.02391	[0.0202, 0.0276]	0.00439	5	0
vanilla	pointer	0.1	0.02357	[0.0205, 0.0274]	0.00433	5	0
vanilla	pointer	1	0.02413	[0.021, 0.027]	0.00368	5	0
vanilla	pointer	10	0.0236	[0.0205, 0.027]	0.00413	5	0
vanilla	pointer	100	0.02454	[0.0205, 0.0286]	0.00495	5	0
vanilla	pointer	1000	0.02415	[0.0207, 0.027]	0.00397	5	0
vanilla	pointer	1e+04	0.02404	[0.0205, 0.0272]	0.00434	5	0
vanilla	seq2seq	0	2.203×10^{-4}	[0, 6.61×10^{-4}]	4.93×10^{-4}	5	0
vanilla	seq2seq	0.01	3.302×10^{-4}	[0, 9.91×10^{-4}]	7.38×10^{-4}	5	0
vanilla	seq2seq	0.1	4.397×10^{-4}	[0, 0.00132]	9.83×10^{-4}	5	0
vanilla	seq2seq	1	8.704×10^{-4}	[0, 0.00219]	0.00143	5	0
vanilla	seq2seq	10	3.303×10^{-4}	[0, 9.91×10^{-4}]	7.39×10^{-4}	5	0
vanilla	seq2seq	100	6.557×10^{-4}	[0, 0.00176]	0.00119	5	0
vanilla	seq2seq	1000	0.001592	[4.23×10^{-4}, 0.00276]	0.00158	5	0
vanilla	seq2seq	1e+04	0.00415	[0.00256, 0.0059]	0.00215	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.002906	[-0.00375, 0.004]	1.0000
adaptive	pointer	0.1	-0.005069	[-0.00716, 0.00378]	1.0000
adaptive	pointer	1	0.003718	[-0.00156, 0.00938]	1.0000
adaptive	pointer	10	0.003214	[-0.00265, 0.00689]	1.0000
adaptive	pointer	100	-0.004713	[-0.00604, 0.00356]	1.0000
adaptive	pointer	1000	0.004852	[-0.0046, 0.00548]	1.0000
adaptive	pointer	1e+04	0.004246	[-0.00313, 0.00531]	1.0000
adaptive	seq2seq	0.01	0	[-6.60×10^{-4}, 0]	1.0000
adaptive	seq2seq	0.1	0	[-3.30×10^{-4}, 0]	1.0000
adaptive	seq2seq	1	0	[-6.60×10^{-4}, 0]	1.0000
adaptive	seq2seq	10	0	[0, 3.31×10^{-4}]	1.0000

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	100	0	$[0, 6.61 \times 10^{-4}]$	1.0000
adaptive	seq2seq	1000	0.002105	$[6.56 \times 10^{-4}, 0.00236]$	0.7500
adaptive	seq2seq	1e+04	0.0048	[0.00309, 0.00603]	0.4375
tropical	pointer	0.01	5.295×10^{-4}	$[5.27 \times 10^{-4}, 0.00324]$	0.4375
tropical	pointer	0.1	0.003258	[0.00255, 0.00388]	0.4375
tropical	pointer	1	0.002144	$[6.17 \times 10^{-6}, 0.00581]$	1.0000
tropical	pointer	10	0.001585	$[-6.90 \times 10^{-6}, 0.00485]$	1.0000
tropical	pointer	100	0.002143	$[1.67 \times 10^{-6}, 0.00344]$	0.9375
tropical	pointer	1000	0.002688	$[5.36 \times 10^{-4}, 0.00442]$	1.0000
tropical	pointer	1e+04	5.536×10^{-4}	$[-8.72 \times 10^{-4}, 0.00138]$	1.0000
tropical	seq2seq	0.01	0	[0, 0.0146]	1.0000
tropical	seq2seq	0.1	0	[0, 0.0019]	1.0000
tropical	seq2seq	1	0	[0, 0.0103]	1.0000
tropical	seq2seq	10	0	[0, 0.0109]	1.0000
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	$[0, 9.50 \times 10^{-4}]$	1.0000
tropical	seq2seq	1e+04	0.003262	[0.00129, 0.00405]	0.7500
vanilla	pointer	0.01	0.003695	[0.00212, 0.00659]	0.4375
vanilla	pointer	0.1	0.00313	[0.00125, 0.00693]	0.4375
vanilla	pointer	1	0.004245	[0.00261, 0.00647]	0.4375
vanilla	pointer	10	0.003649	[0.00178, 0.00627]	0.4375
vanilla	pointer	100	0.004802	[0.00261, 0.00777]	0.4375
vanilla	pointer	1000	0.004843	[0.00261, 0.00614]	0.4375
vanilla	pointer	1e+04	0.004641	[0.00239, 0.00627]	0.4375
vanilla	seq2seq	0.01	0	$[0, 3.30 \times 10^{-4}]$	1.0000
vanilla	seq2seq	0.1	0	$[0, 6.58 \times 10^{-4}]$	1.0000
vanilla	seq2seq	1	0	[0, 0.00151]	1.0000
vanilla	seq2seq	10	0	$[0, 3.30 \times 10^{-4}]$	1.0000
vanilla	seq2seq	100	0	$[0, 9.88 \times 10^{-4}]$	1.0000
vanilla	seq2seq	1000	0.002114	$[-2.20 \times 10^{-4}, 0.00276]$	1.0000
vanilla	seq2seq	1e+04	0.003245	[0.00256, 0.00568]	0.4375

Table 27: `id_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.01556	[-0.21, 0.0951]	0.9293	-0.019	0.8946
adaptive	seq2seq	0.7135	[0.663, 0.799]	$< 10^{-4}$	0.484	0.0002
tropical	pointer	-0.1231	[-0.24, -0.00556]	0.4813	-0.0895	0.5151
tropical	seq2seq	0.1606	[-0.172, 0.538]	0.3566	0.112	0.3039
vanilla	pointer	0.1182	[-0.209, 0.466]	0.4990	0.0857	0.5387
vanilla	seq2seq	0.519	[0.303, 0.793]	0.0014	0.37	0.0020

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.01615	[-0.0206, -0.0117]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.01895	[-0.0253, -0.0126]	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.01692	[-0.0222, -0.0116]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.01776	[-0.0258, -0.00971]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.0181	[-0.0264, -0.00986]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.01925	[-0.028, -0.0105]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.01821	[-0.0244, -0.0121]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.02016	[-0.0252, -0.0151]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-2.203×10^{-4}	$[-8.32 \times 10^{-4}, 3.91 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.004535	[-0.00923, 0.0183]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	1.931×10^{-4}	[-0.00218, 0.00257]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.003123	[-0.00552, 0.0118]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.003437	[-0.00658, 0.0135]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-6.557×10^{-4}	$[-0.00214, 8.25 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.001276	$[-0.00272, 1.71 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.001476	$[-0.00376, 8.11 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.001301	[-0.0151, 0.0125]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.005375	[-0.012, 0.00127]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.007114	$[-0.0133, -9.30 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.002048	[-0.0141, 0.00996]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.003273	[-0.0137, 0.00717]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.007496	[-0.0131, -0.00185]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.005417	[-0.0137, 0.00288]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.004197	[-0.014, 0.00564]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-1.212×10^{-7}	$[-4.58 \times 10^{-7}, 2.15 \times 10^{-7}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-3.302×10^{-4}	$[-0.00125, 5.87 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-3.295×10^{-4}	$[-0.00124, 5.85 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-8.704×10^{-4}	$[-0.00265, 9.06 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	9.094×10^{-8}	$[-1.62 \times 10^{-7}, 3.43 \times 10^{-7}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-2.153×10^{-4}	$[-5.81 \times 10^{-4}, 1.51 \times 10^{-4}]$	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	2.331×10^{-4}	[-0.00213, 0.0026]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	6.362×10^{-4}	$[-2.23 \times 10^{-4}, 0.0015]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01808	[0.00642, 0.0297]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01853	[0.00894, 0.0281]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01635	[0.0117, 0.021]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.02208	[0.00724, 0.0369]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01999	[0.0059, 0.0341]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0166	[0.0112, 0.022]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0169	[0.00626, 0.0275]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01505	[0.00383, 0.0263]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.003455	$[4.10 \times 10^{-4}, 0.0065]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	9.427×10^{-5}	[-0.0156, 0.0158]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.006019	[0.00224, 0.00979]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.002372	[-0.0112, 0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.001726	[-0.0115, 0.015]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.005284	$[-1.94 \times 10^{-4}, 0.0108]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.005618	$[-6.48 \times 10^{-4}, 0.0119]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0012	$[-9.01 \times 10^{-4}, 0.0033]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01938	[0.0143, 0.0245]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.02358	[0.0181, 0.0291]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.02313	[0.0174, 0.0289]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02326	[0.019, 0.0276]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.02327	[0.0179, 0.0287]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02388	[0.0175, 0.0303]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.02255	[0.0176, 0.0275]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01989	[0.0169, 0.0229]	1.0000

A.7 id_exact

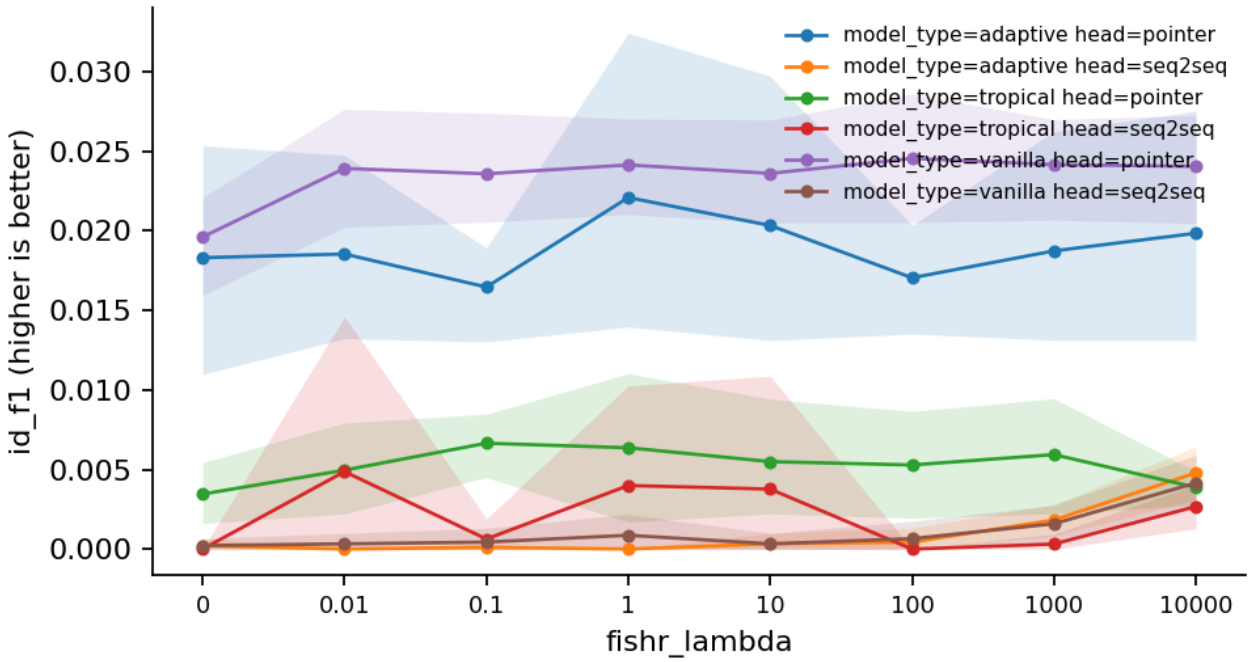


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

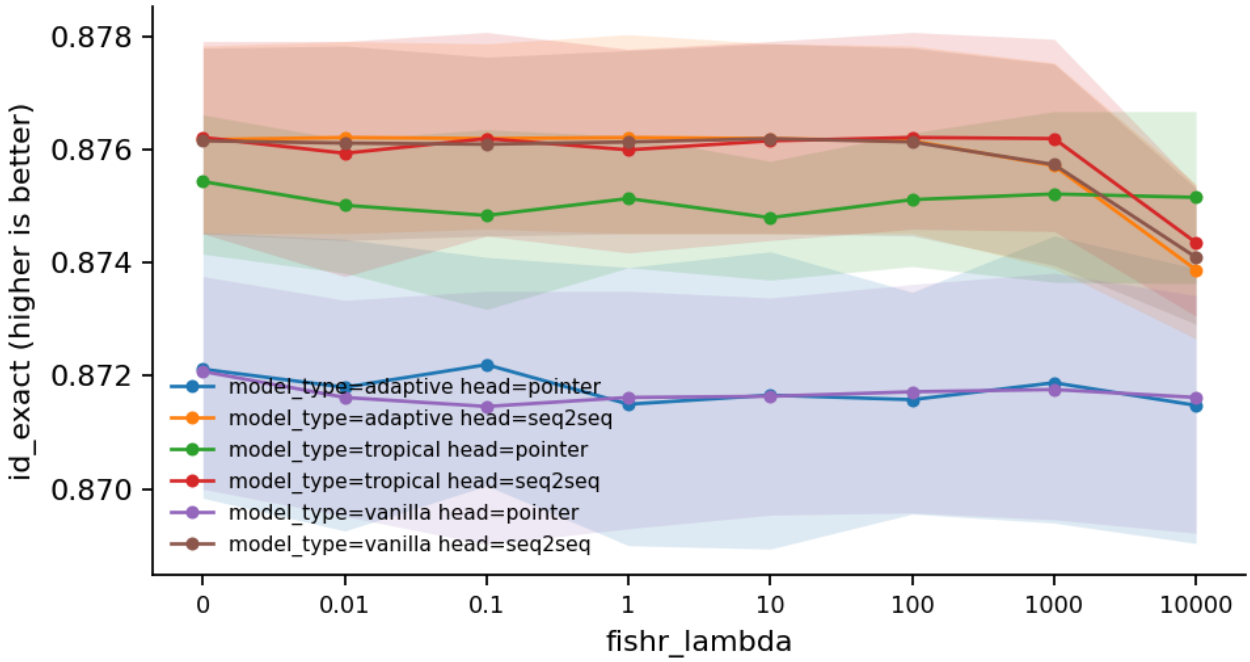


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8721	[0.87, 0.875]	0.0031	5	0
adaptive	pointer	0.01	0.8718	[0.869, 0.874]	0.00332	5	0
adaptive	pointer	0.1	0.8722	[0.87, 0.874]	0.00278	5	0
adaptive	pointer	1	0.8715	[0.869, 0.874]	0.00331	5	0
adaptive	pointer	10	0.8717	[0.869, 0.874]	0.00341	5	0
adaptive	pointer	100	0.8716	[0.87, 0.873]	0.00261	5	0
adaptive	pointer	1000	0.8719	[0.869, 0.874]	0.00338	5	0
adaptive	pointer	1e+04	0.8715	[0.869, 0.874]	0.00328	5	0
adaptive	seq2seq	0	0.8762	[0.875, 0.878]	0.00216	5	0
adaptive	seq2seq	0.01	0.8762	[0.875, 0.878]	0.00222	5	0
adaptive	seq2seq	0.1	0.8762	[0.875, 0.878]	0.00219	5	0
adaptive	seq2seq	1	0.8762	[0.875, 0.878]	0.00222	5	0
adaptive	seq2seq	10	0.8762	[0.875, 0.878]	0.00219	5	0
adaptive	seq2seq	100	0.8762	[0.875, 0.878]	0.00218	5	0
adaptive	seq2seq	1000	0.8757	[0.874, 0.878]	0.00241	5	0
adaptive	seq2seq	1e+04	0.8739	[0.873, 0.875]	0.00172	5	0
tropical	pointer	0	0.8754	[0.874, 0.877]	0.0016	5	0
tropical	pointer	0.01	0.875	[0.874, 0.876]	0.00152	5	0
tropical	pointer	0.1	0.8748	[0.873, 0.876]	0.00202	5	0
tropical	pointer	1	0.8751	[0.874, 0.876]	0.00147	5	0
tropical	pointer	10	0.8748	[0.874, 0.876]	0.00135	5	0
tropical	pointer	100	0.8751	[0.874, 0.876]	0.00157	5	0
tropical	pointer	1000	0.8752	[0.874, 0.877]	0.00194	5	0
tropical	pointer	1e+04	0.8752	[0.874, 0.877]	0.00201	5	0
tropical	seq2seq	0	0.8762	[0.875, 0.878]	0.00222	5	0
tropical	seq2seq	0.01	0.8759	[0.874, 0.878]	0.00265	5	0
tropical	seq2seq	0.1	0.8762	[0.874, 0.878]	0.00225	5	0
tropical	seq2seq	1	0.876	[0.874, 0.878]	0.00238	5	0
tropical	seq2seq	10	0.8762	[0.874, 0.878]	0.00231	5	0
tropical	seq2seq	100	0.8762	[0.875, 0.878]	0.00222	5	0
tropical	seq2seq	1000	0.8762	[0.875, 0.878]	0.00217	5	0
tropical	seq2seq	1e+04	0.8744	[0.873, 0.875]	0.00149	5	0
vanilla	pointer	0	0.8721	[0.87, 0.874]	0.00246	5	0
vanilla	pointer	0.01	0.8716	[0.87, 0.873]	0.00254	5	0
vanilla	pointer	0.1	0.8715	[0.869, 0.874]	0.00289	5	0
vanilla	pointer	1	0.8716	[0.869, 0.873]	0.0027	5	0
vanilla	pointer	10	0.8716	[0.87, 0.873]	0.00248	5	0
vanilla	pointer	100	0.8717	[0.87, 0.874]	0.00271	5	0
vanilla	pointer	1000	0.8718	[0.869, 0.874]	0.0029	5	0
vanilla	pointer	1e+04	0.8716	[0.869, 0.873]	0.0027	5	0
vanilla	seq2seq	0	0.8762	[0.875, 0.878]	0.00213	5	0
vanilla	seq2seq	0.01	0.8761	[0.874, 0.878]	0.00224	5	0
vanilla	seq2seq	0.1	0.8761	[0.874, 0.878]	0.0021	5	0
vanilla	seq2seq	1	0.8761	[0.875, 0.878]	0.0021	5	0
vanilla	seq2seq	10	0.8762	[0.875, 0.878]	0.00219	5	0
vanilla	seq2seq	100	0.8761	[0.874, 0.878]	0.00215	5	0
vanilla	seq2seq	1000	0.8757	[0.874, 0.878]	0.00237	5	0
vanilla	seq2seq	1e+04	0.8741	[0.873, 0.875]	0.00166	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-3.200×10^{-4}	[-0.00584, 0.0052]	1.0000
adaptive	pointer	0.1	8.000×10^{-5}	[-0.00544, 0.0056]	1.0000
adaptive	pointer	1	-6.200×10^{-4}	[-0.00614, 0.0049]	0.9998
adaptive	pointer	10	-4.600×10^{-4}	[-0.00598, 0.00506]	1.0000
adaptive	pointer	100	-5.400×10^{-4}	[-0.00606, 0.00498]	0.9999
adaptive	pointer	1000	-2.400×10^{-4}	[-0.00576, 0.00528]	1.0000
adaptive	pointer	1e+04	-6.400×10^{-4}	[-0.00616, 0.00488]	0.9997
adaptive	seq2seq	0.01	4.000×10^{-5}	[-0.00375, 0.00383]	1.0000
adaptive	seq2seq	0.1	2.000×10^{-5}	[-0.00377, 0.00381]	1.0000
adaptive	seq2seq	1	4.000×10^{-5}	[-0.00375, 0.00383]	1.0000
adaptive	seq2seq	10	2.000×10^{-5}	[-0.00377, 0.00381]	1.0000
adaptive	seq2seq	100	-2.000×10^{-5}	[-0.00381, 0.00377]	1.0000
adaptive	seq2seq	1000	-4.600×10^{-4}	[-0.00425, 0.00333]	0.9996
adaptive	seq2seq	1e+04	-0.0023	[-0.00609, 0.00149]	0.4043
tropical	pointer	0.01	-4.200×10^{-4}	[-0.0034, 0.00256]	0.9990
tropical	pointer	0.1	-6.000×10^{-4}	[-0.00358, 0.00238]	0.9913
tropical	pointer	1	-3.000×10^{-4}	[-0.00328, 0.00268]	0.9999
tropical	pointer	10	-6.400×10^{-4}	[-0.00362, 0.00234]	0.9876
tropical	pointer	100	-3.200×10^{-4}	[-0.0033, 0.00266]	0.9998
tropical	pointer	1000	-2.200×10^{-4}	[-0.0032, 0.00276]	1.0000
tropical	pointer	1e+04	-2.800×10^{-4}	[-0.00326, 0.0027]	0.9999
tropical	seq2seq	0.01	-2.800×10^{-4}	[-0.00418, 0.00362]	1.0000
tropical	seq2seq	0.1	-2.000×10^{-5}	[-0.00392, 0.00388]	1.0000
tropical	seq2seq	1	-2.200×10^{-4}	[-0.00412, 0.00368]	1.0000
tropical	seq2seq	10	-6.000×10^{-5}	[-0.00396, 0.00384]	1.0000
tropical	seq2seq	100	0	[-0.0039, 0.0039]	1.0000
tropical	seq2seq	1000	-2.000×10^{-5}	[-0.00392, 0.00388]	1.0000
tropical	seq2seq	1e+04	-0.00186	[-0.00576, 0.00204]	0.6470
vanilla	pointer	0.01	-4.600×10^{-4}	[-0.00514, 0.00422]	0.9999
vanilla	pointer	0.1	-6.200×10^{-4}	[-0.0053, 0.00406]	0.9993
vanilla	pointer	1	-4.600×10^{-4}	[-0.00514, 0.00422]	0.9999
vanilla	pointer	10	-4.400×10^{-4}	[-0.00512, 0.00424]	0.9999
vanilla	pointer	100	-3.600×10^{-4}	[-0.00504, 0.00432]	1.0000
vanilla	pointer	1000	-3.200×10^{-4}	[-0.005, 0.00436]	1.0000
vanilla	pointer	1e+04	-4.600×10^{-4}	[-0.00514, 0.00422]	0.9999
vanilla	seq2seq	0.01	-4.000×10^{-5}	[-0.00375, 0.00367]	1.0000
vanilla	seq2seq	0.1	-6.000×10^{-5}	[-0.00377, 0.00365]	1.0000
vanilla	seq2seq	1	-2.000×10^{-5}	[-0.00373, 0.00369]	1.0000
vanilla	seq2seq	10	4.000×10^{-5}	[-0.00367, 0.00375]	1.0000
vanilla	seq2seq	100	-2.000×10^{-5}	[-0.00373, 0.00369]	1.0000
vanilla	seq2seq	1000	-4.200×10^{-4}	[-0.00413, 0.00329]	0.9998
vanilla	seq2seq	1e+04	-0.00206	[-0.00577, 0.00165]	0.4960

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-5.429×10^{-5}	$[-1.08 \times 10^{-4}, -1.29 \times 10^{-5}]$	0.8276	-0.0324	0.8084
adaptive	seq2seq	-2.871×10^{-4}	$[-4.08 \times 10^{-4}, -1.92 \times 10^{-4}]$	0.1121	-0.253	0.0542
tropical	pointer	4.143×10^{-5}	$[-4.07 \times 10^{-5}, 1.51 \times 10^{-4}]$	0.7594	0.021	0.8832
tropical	seq2seq	-1.614×10^{-4}	$[-2.89 \times 10^{-4}, -3.36 \times 10^{-5}]$	0.3766	-0.112	0.3903
vanilla	pointer	2.500×10^{-5}	$[-1.71 \times 10^{-5}, 7.07 \times 10^{-5}]$	0.9064	0.00952	0.9498

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-2.421×10^{-4}	$[-3.44 \times 10^{-4}, -1.53 \times 10^{-4}]$	0.1668	-0.211	0.1186

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00336	[0.00196, 0.00476]	0.0373	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0034	[0.00194, 0.00486]	0.0384	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.00338	[0.00215, 0.00461]	0.0251	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.00352	[0.00179, 0.00525]	0.0532	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.00316	[0.00146, 0.00486]	0.0668	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0034	[0.0014, 0.0054]	0.0829	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.00346	[0.00184, 0.00508]	0.0487	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.00354	[0.00214, 0.00494]	0.0328	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	6.000×10^{-5}	$[-1.07 \times 10^{-4}, 2.27 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-1.800×10^{-4}	$[-8.28 \times 10^{-4}, 4.68 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	1.000×10^{-4}	$[-1.78 \times 10^{-4}, 3.78 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-1.400×10^{-4}	$[-6.49 \times 10^{-4}, 3.69 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-4.000×10^{-5}	$[-2.28 \times 10^{-4}, 1.48 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	8.000×10^{-5}	$[-8.19 \times 10^{-5}, 2.42 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	4.600×10^{-4}	$[1.74 \times 10^{-4}, 7.46 \times 10^{-4}]$	0.0887	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	2.600×10^{-4}	[-0.00131, 0.00183]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	4.000×10^{-5}	[-0.00248, 0.00256]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	1.800×10^{-4}	[-0.00191, 0.00227]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	7.400×10^{-4}	$[-4.91 \times 10^{-4}, 0.00197]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-1.200×10^{-4}	[-0.00225, 0.00201]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	2.000×10^{-5}	[-0.00199, 0.00203]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-1.400×10^{-4}	[-0.00191, 0.00163]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	1.200×10^{-4}	[-0.00158, 0.00182]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-1.400×10^{-4}	[-0.00222, 0.00194]	1.0000	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	2.000×10^{-5}	$[-3.55 \times 10^{-5}, 7.55 \times 10^{-5}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	1.000×10^{-4}	$[-7.56 \times 10^{-5}, 2.76 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	1.000×10^{-4}	$[-1.15 \times 10^{-4}, 3.15 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	8.000×10^{-5}	$[-1.42 \times 10^{-4}, 3.02 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	2.000×10^{-5}	$[-3.55 \times 10^{-5}, 7.55 \times 10^{-5}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-2.000×10^{-5}	$[-1.82 \times 10^{-4}, 1.42 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-2.200×10^{-4}	$[-6.62 \times 10^{-4}, 2.22 \times 10^{-4}]$	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.00406	[-0.00584, -0.00228]	0.0481	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00442	[-0.00648, -0.00236]	0.0556	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.004	[-0.00533, -0.00267]	0.0235	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.00472	[-0.00709, -0.00235]	0.0608	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.00454	[-0.00676, -0.00232]	0.0608	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.00458	[-0.00625, -0.00291]	0.0285	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.00384	[-0.00571, -0.00197]	0.0608	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.0024	[-0.00599, 0.00119]	0.5285	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-7.800×10^{-4}	$[-0.00182, 2.57 \times 10^{-4}]$	0.5247	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-9.200×10^{-4}	$[-0.0026, 7.63 \times 10^{-4}]$	0.5285	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.00136	$[-0.00221, -5.06 \times 10^{-4}]$	0.1150	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-8.600×10^{-4}	$[-0.00213, 4.05 \times 10^{-4}]$	0.5285	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.00136	$[-0.00295, 2.29 \times 10^{-4}]$	0.4579	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.0011	$[-0.00211, -9.13 \times 10^{-5}]$	0.2721	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-9.800×10^{-4}	$[-0.00177, -1.88 \times 10^{-4}]$	0.2335	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	8.000×10^{-4}	$[-6.24 \times 10^{-4}, 0.00222]$	0.5285	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.00408	[-0.00558, -0.00258]	0.0285	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0045	[-0.00574, -0.00326]	0.0130	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.00464	[-0.00623, -0.00305]	0.0239	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.00452	[-0.00604, -0.003]	0.0236	*

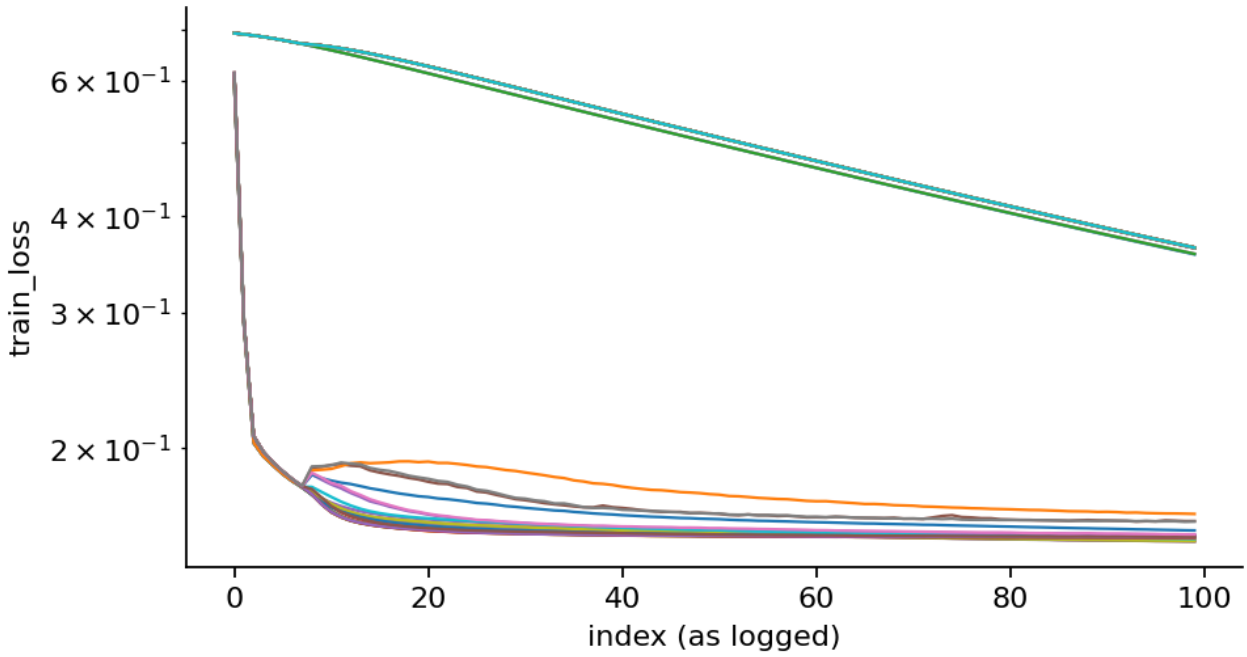


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.00456	[-0.00593, -0.00319]	0.0175	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.00442	[-0.00584, -0.003]	0.0217	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.00398	[-0.00551, -0.00245]	0.0315	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.00248	[-0.00447, -4.87×10^{-4}]	0.2335	

B Curves

C Configuration

The base configuration every run shares (frozen hash `096095567af4`); factor levels are in Table 1.

```
task:
  name: ThreeSumDecisionDataset
  value_range:
  - -20
  - 20
  target_range:
  - -75
  - 75
  weight_range: null
  adversarial_range:
  - 40
  - 60
  noise_prob: 0.0
```

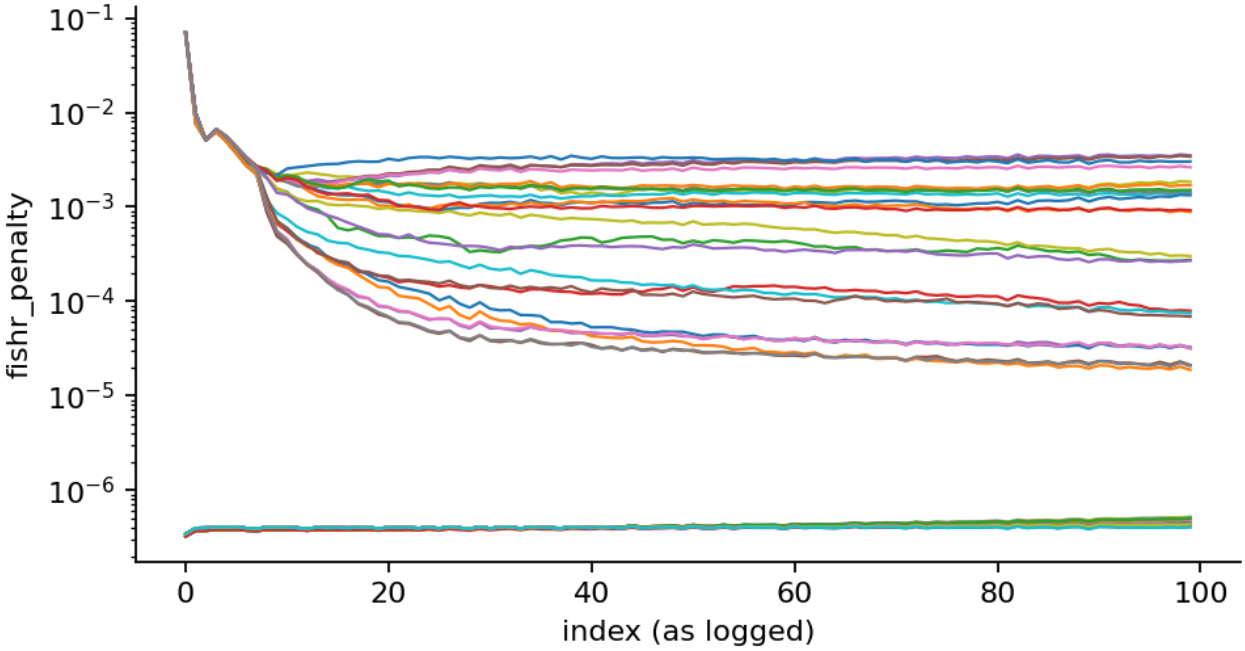


Figure 10: `fishr_penalty`, the mean over seeds per cell.

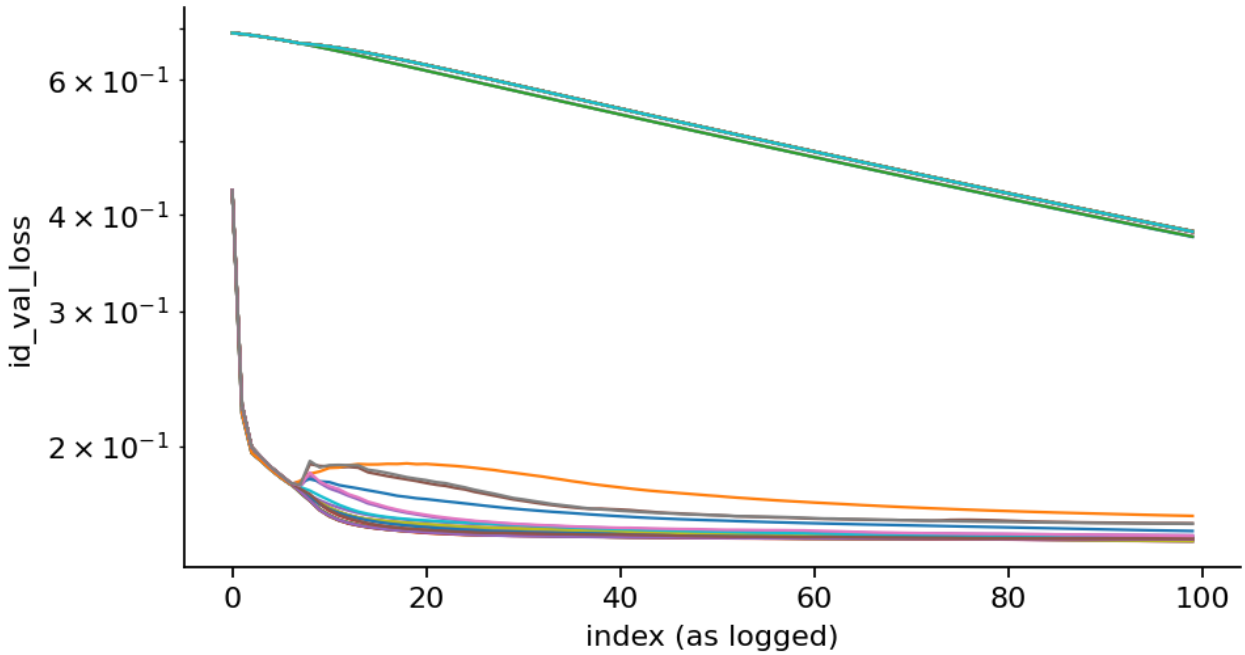


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - -375
  - 375
  ood_noise_range:
  - 40
  - 60
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/ThreeSumDecision/results`.